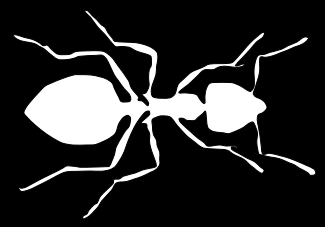
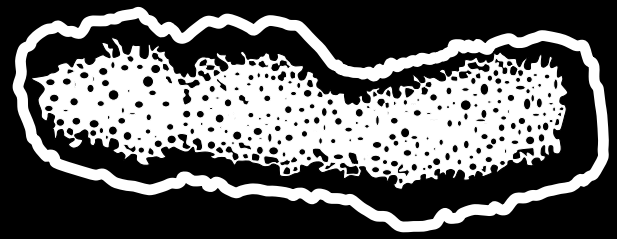
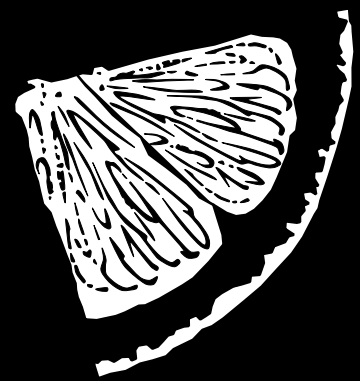
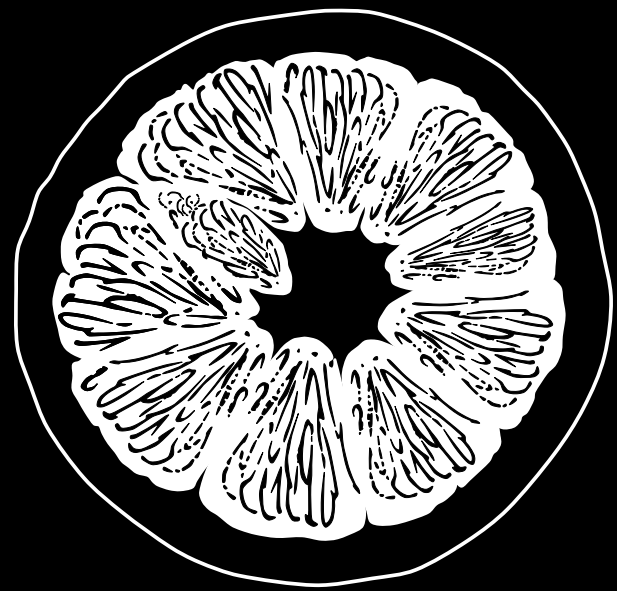
SURFACES & SHAPES

ISABELLA HANDBURY
2024 SEM 2 FODR



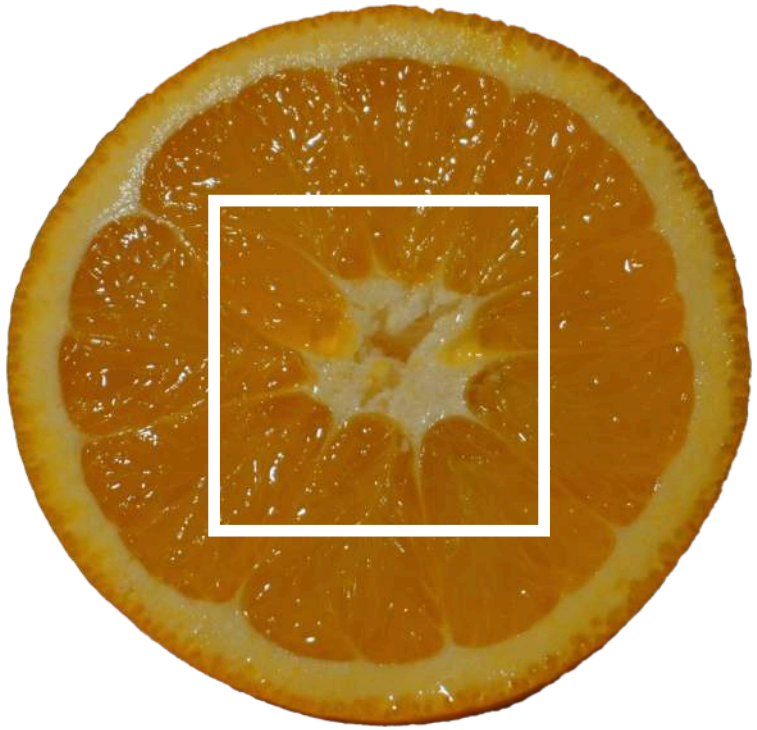






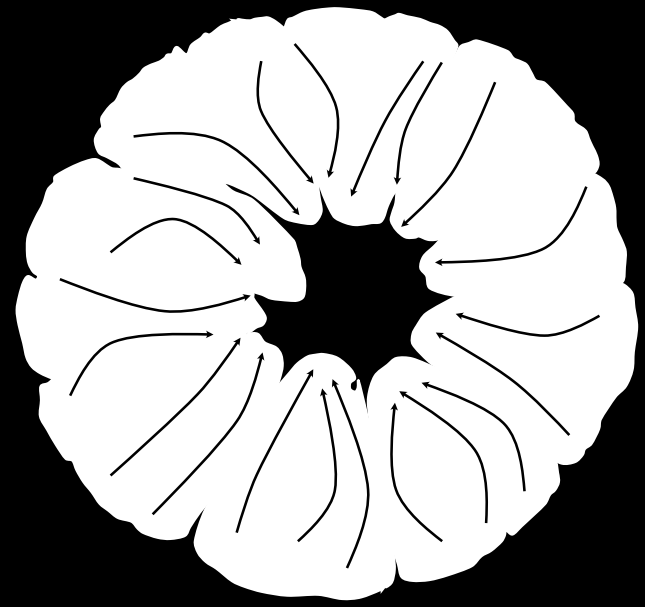
CITRUS X SINENSIS

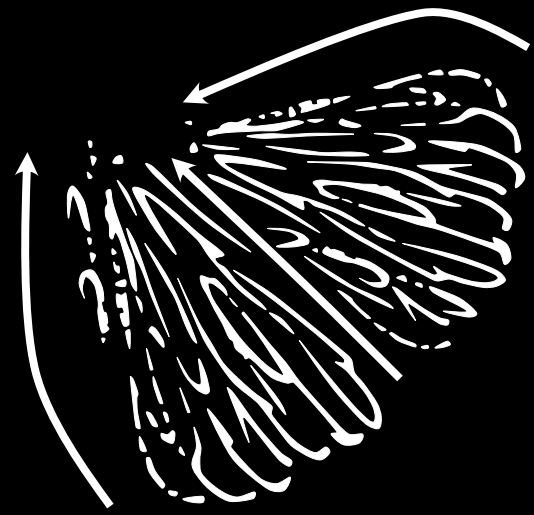




THE CENTRAL COLUMN OF THE ORANGE IS FORMED AS A SUPPORT STRUCTURE BOTH DURING THE ORANGE'S GROWTH PROCESS AS WELL AS TO IMPROVE STRUCTURAL INTERGITY WITH THE RIPE FRUIT.

THIS RESULTS IN THE PATTERN OF TENSION BETWEEN THE CARPELS (JUICE STORAGE) AND THE CENTRAL PITH COLUMN. THIS PATTERN DRAWING TOWARDS A CONCAVE CENTER IN CONJUNTION WITH THE PATTERNATION OF THE JUICE STORAGE FORMED THROUGH GROWTH OF JUICE FOLLICLES PRODUCES UNIQUE SURFACE CHARACTERISTICS.

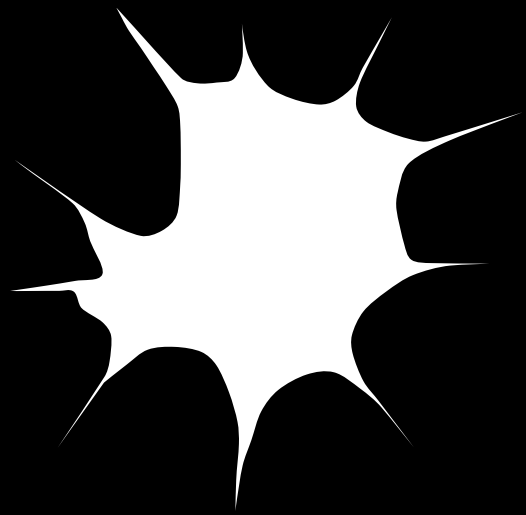




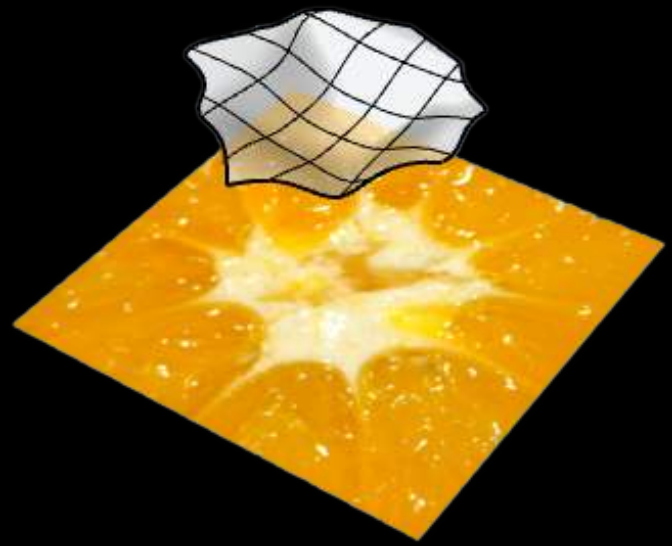
RAISED CENTRAL COLUMN IS A CONJUNTION OF BOTH A RAISED COMPARATIVE SURFACE AND A SUNKEN ELEMENT

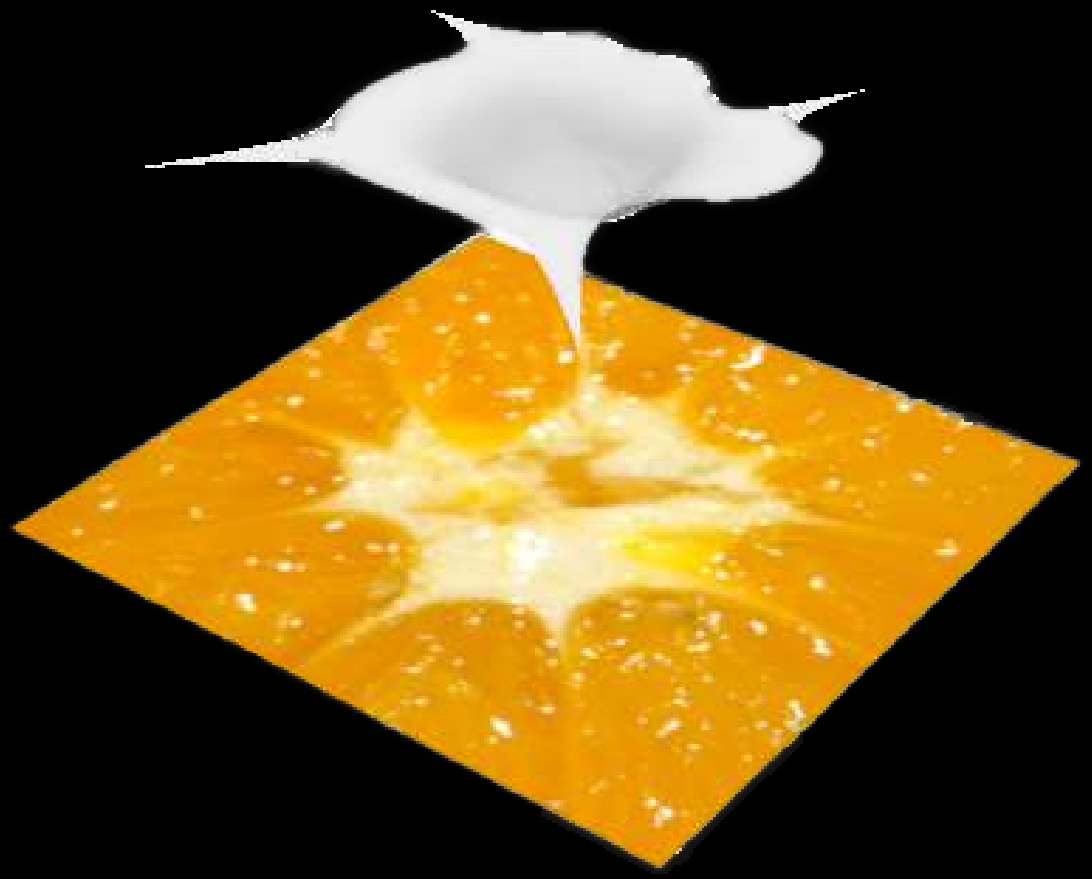
PATTERNISTIC EXTENSION OF THE RAISED PITH INTO THE CARPELS

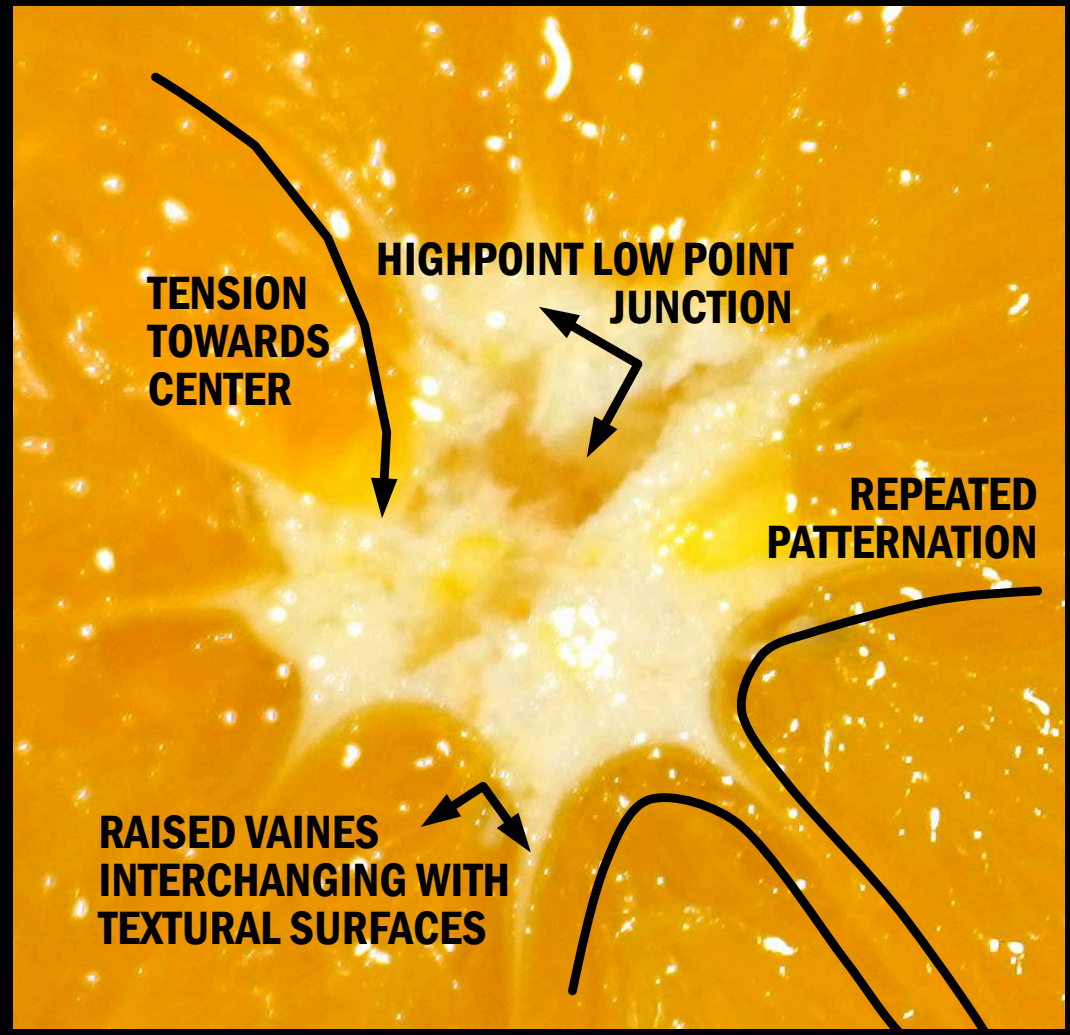




TEXTURAL DIFFERENCES WITH THE JUICE STORAGE PEAKS AND TROUGHS DUE TO THE TENSION OF THE SKIN






TENSION
TOWARDS
CENTER
HIGHPOINT LOW POINT
JUNCTION
REPEATED
PATTERNATION
RAISED VAINES
INTERCHANGING WITH
TEXTURAL SURFACES

BIOMIMETIC DESIGN

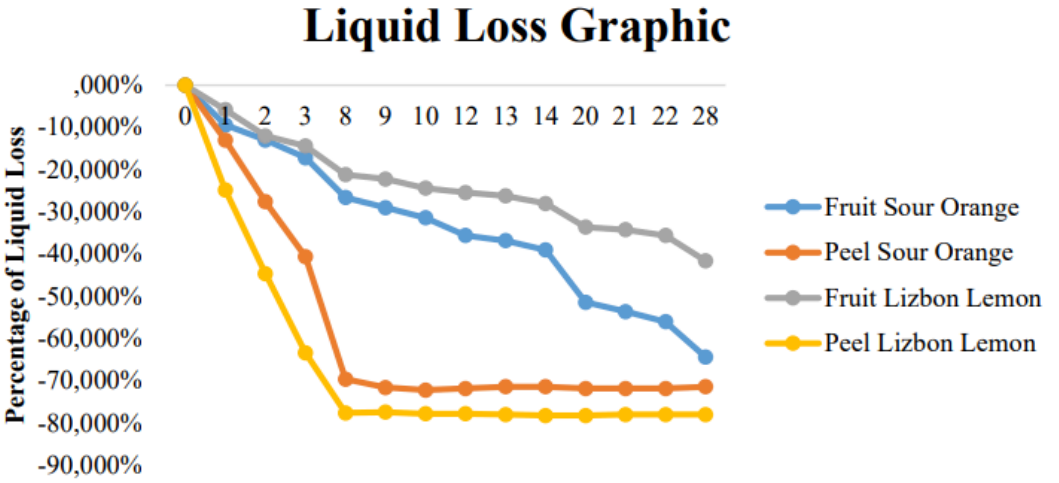
BIOMIMETIC DESIGN IS AN EMERGING TOOL IN A DESIGNER'S TOOL BOX. IT FACILITATES A LEARNING PATTERN STEMMING FROM NATURAL SPECIES WHO HAVE OPTEMISED THEIR FEATURES FOR SURVIVAL.

AN EXAMPLE OF BIOMIMICRY INVOLVING ORANGES IS A MASTER'S THESIS PUBLISHED IN 2022 AT THE IZMIR UNIVERSITY OF ECONOMICS. THE THESIS TITLED ' LESSONS FROM CITRUS PEEL: A BIOMIMETIC APPROACH FOR PACKAGING DESIGN' DISCUSSES A NEW APPROACH TO PACKAGING MATERIALS AND DESIGN THOUGH THE LENS OF CITRUS PEELS' BIOMETRIC PRINCIPALS.

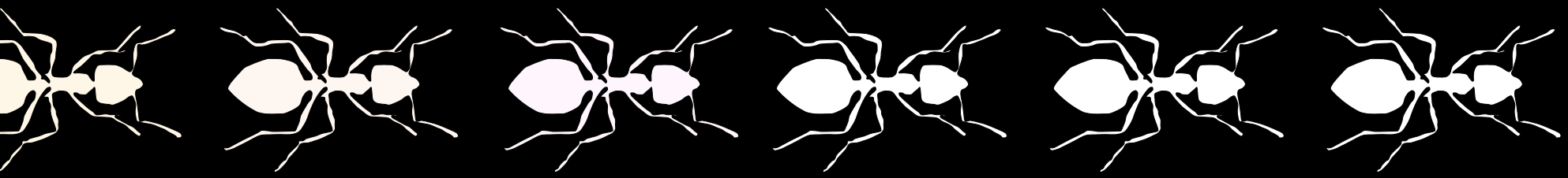
AS MENTIONED IN THE THESIS " PLANTS HAVE EVOLVED AN ARSENAL OF SOLUTIONS TO MAKE THEM SUITABLE FOR LIFE,' I BELIEVE THIS IS THE ESSENCE OF BIOMETRIC DESIGN WHERE WE ADAPT THE EVOLUTIONARY PERFECTED PATTERNS AND SURFACES OF PLANTS TO FIT OUR DESIGN NEEDS.

THE THESIS GOES ON TO EXPLORE EXAMPLES OF CITRUS IN BIOMIMETRIC DESIGN INCLUDING CITRUS LEATHER, PACKAGING AND FABRIC. THE CITRUS PEEL'S ABILITY TO RETAIN MOISTURE AND NEGATE FLUID LOSS IS HIGHLIGHTED AS A KEY FACTOR EXLORED FOR UTELISING ITS PATTERNS FOR DESIGN.





UNDERSTAND

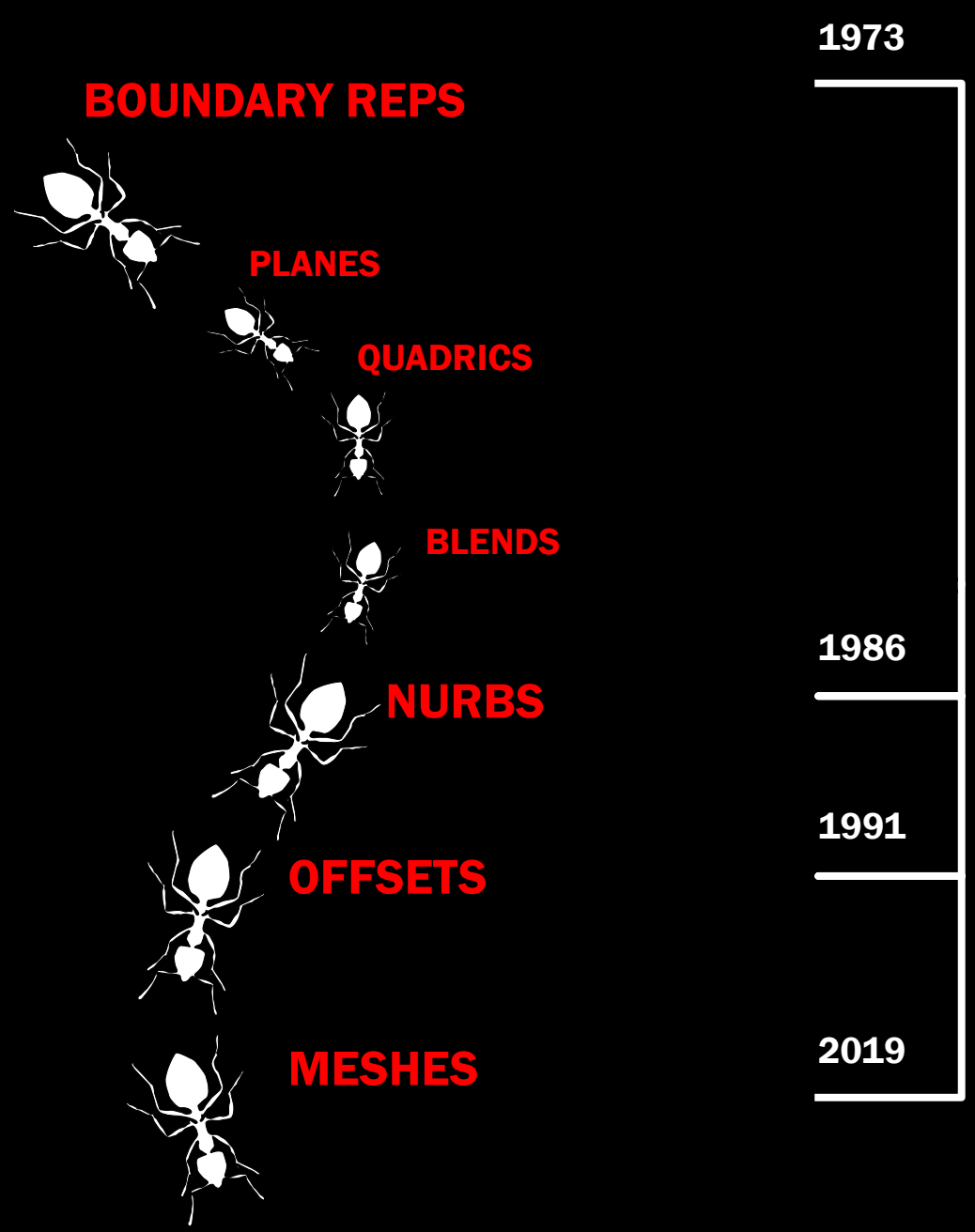


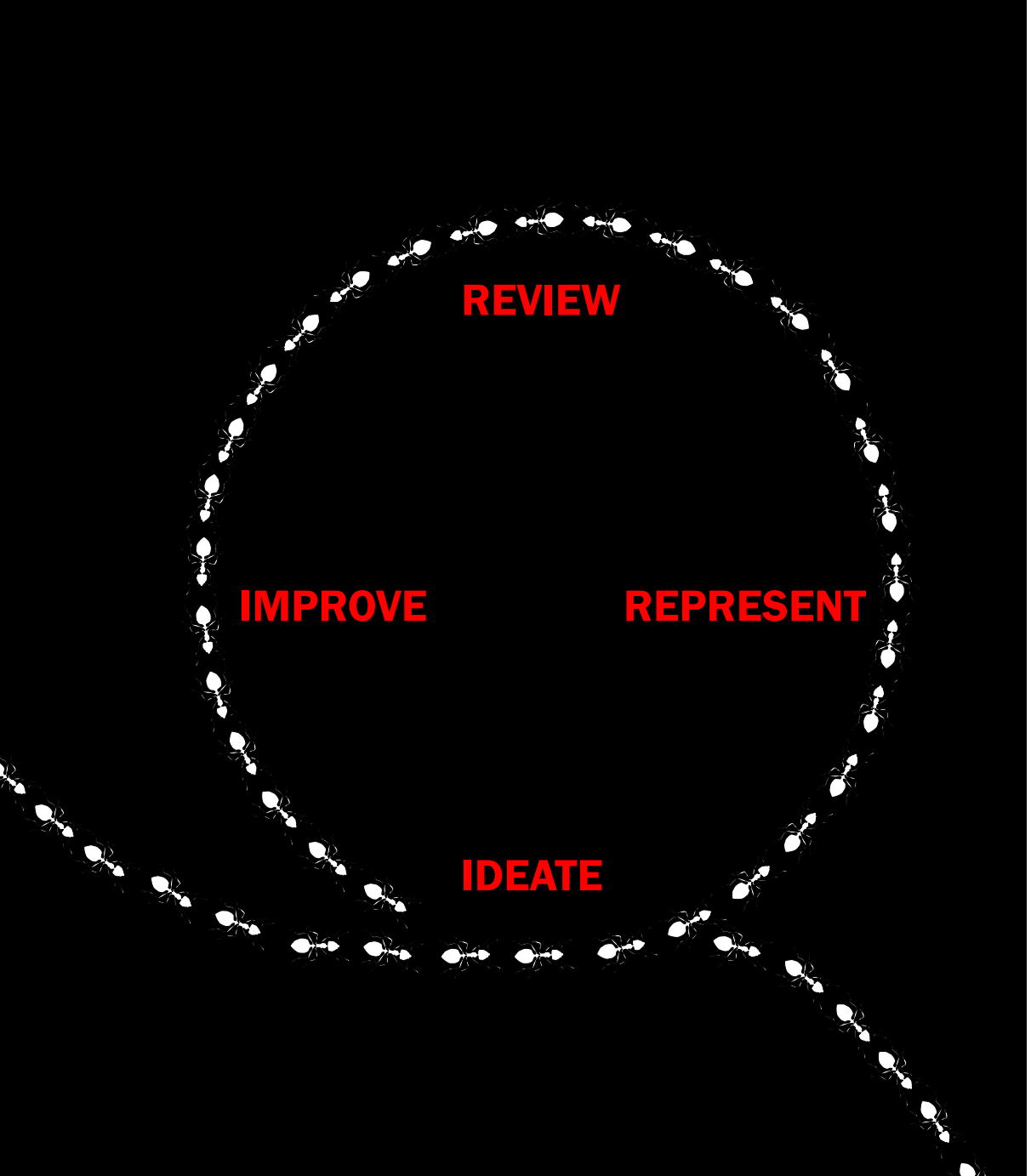
COMPUTATION

COMPUTATIONAL DESIGN EXTENDS FAR BEYOND ITS BASE ABILITIES TO PROCESS HIGH VOLUMES OF DATA, ENDLESSLY REPEAT TASKS AND PROVIDE A COMMUNICATION PLATFORM. AS ADDRESSED IN THE VIDEO BY DIGITAL DESIGN UNIT - TU DARMSTADT THAT WAS PRESENTED IN THE LECUTRES, COMPUTATIONAL DESIGN IS ' THE USE OF RULES AND OPERATIONS TO ORGANISE SPACE, DEFINE FORM AND EVALUATE VARIATIONS '.

BECAUSE OF THESE FUNCTIONALITIES, COMPUTATIONAL DESIGN IS THE MODERN PIONEER OF HUMAN/COMPUTER COLLABORATION, LIKENED IN THE ENGINEERING DISCIPLINE AS ' ENGINEERING AT THE SPEED OF LIGHT' IN BRADLEY ROTHENBERG'S ARTICLE ON ' THE COMPUTATIONAL DESIGN ERA'.

AS DEMONSTRAITED IN THIS DIAGRAM, THE MOVE TOWARDS COMPUTATIONAL DESIGN IS BUILDING OFF OF PRIMARY FORMS OF COMPUTER AIDED DESIGN. HOWEVER, COMPUTATIONAL DESIGN IS MORE PARAMETER DRIVEN, OPTIMISING DESIGNER INPUT RATHER THEN JUST REPRESENTING IT.





ITERATIVE DESIGN

ITERATIVE DESIGN PUT SIMPLISTICALLY IS THE PROCESS OF REPEATED DESIGN WITH IMPROVEMENTS TO EACH VERSION. DEFINED BY THE WORLD DESIGN FOUNDATION AS " A DESIGN METHODOLOGY BASED ON A CYCLIC PROCESS OF PROTOTYPING, TESTING, ANALYSING, AND REFINING A PRODUCT OR PROCESS. BASED ON THE RESULTS OF TESTING THE MOST RECENT ITERATION OF A DESIGN, CHANGES AND REFINEMENTS ARE MADE." THE MOST CRITICAL ASPECT OF THIS DEFINITION IS THE CONNECTION BETWEEN RESULTS AND REFINEMENTS. I HAVE A PREFERENCE TO REFER TO ITERATIVE DESIGN AS RAPID PROTOTYPING , AS THIS TITLE MORE ACCUTELY HIGHLIGHTS THIS VITAL CONNECTION.

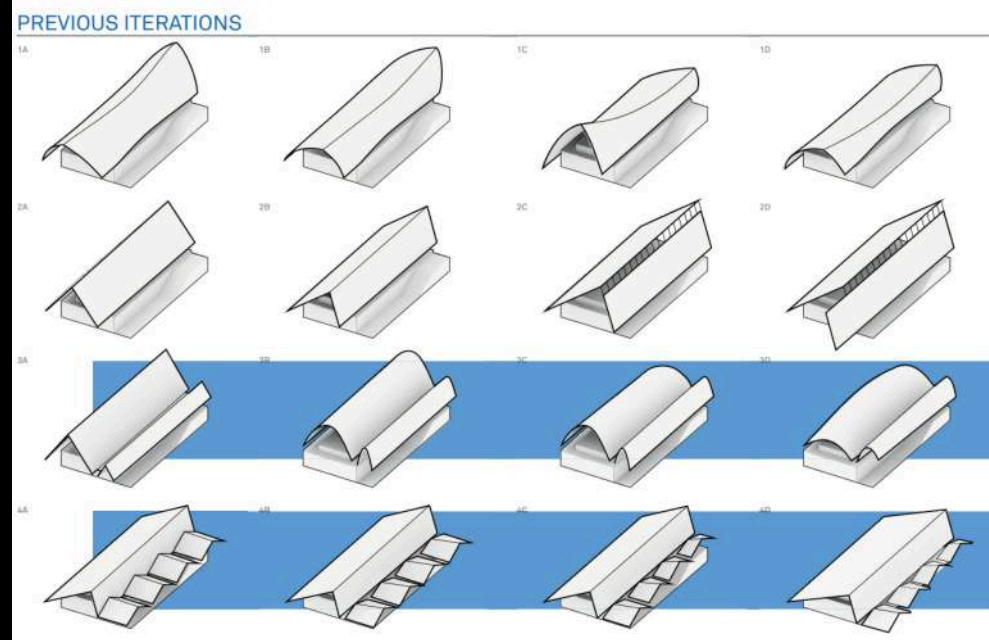
ITERATIVE DESIGN IS ONLY FUNCTIONAL AND BENEFITICAL IF THE NEXT ITERATION IS BUILT UPON IMPROVEMENTS ON A PREVIOUS VERSION. THESE AREAS OF IMPROVEMENT ARE GENERATED THROUGH TESTING, ANALYSIS, CONVERSATIONS WITH STAKEHOLDERS AND CURIOUSITY IN THE BOUNDS OF A DESIGN.

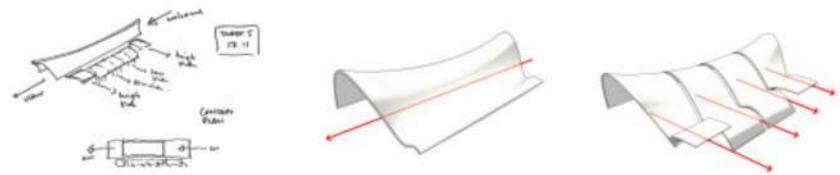
REASONS FOR ITERATION

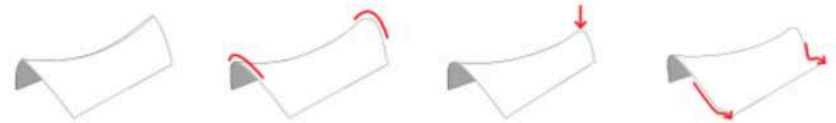
A LOT CAN BE LEARNT FROM THE PROCESSES DISCUSSED BY SONYA PARTON'S LECTURE ON HER PROJECTS AND JOURNEY WITHIN HER DISCIPLINE OF ARCHITECTURE. A KEY TAKE AWAY FOR ME ARE THE REASONS FOR ITERATION IN THE CONTEXT OF A LARGE SCALE DESIGN WITH MAJOR STAKEHOLDERS AND AN EXTENSIVE TEAM.

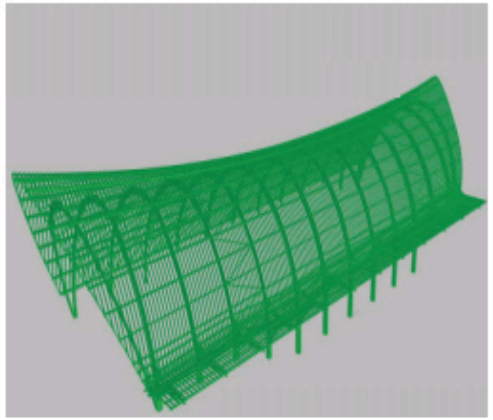
THE ITERATIONS FOR THE ROOF DESIGN OF KUTUBU CONVENTON CENTRE EXPLICITLY DEMONSTRATE THE PURPOSE OF ITERATION TO COMMUNICATE A PATH THROUGH DESIGN AND TO IMPROVE AND CHANGE ELEMENTS OF A DESIGN REPEATEDLY. AS PARTON ADDRESSED " IT [ITERATIONS] WERE GRADUALLY FORMED BY SPEAKING WITH THE CULTURAL CONSULTANT... I THINK WE WENT THROUGH MAYBE 30 OR 40 ROOF ITERATIONS," POSING A REASON FOR WHY WE ITERATE BEING IN ORDER TO PROGRESS THROUGH DISCUSSION WITH INVOLVED PARTIES TOWARDS A DESIGN THAT REPRESENTS THE INTENDED VISION.

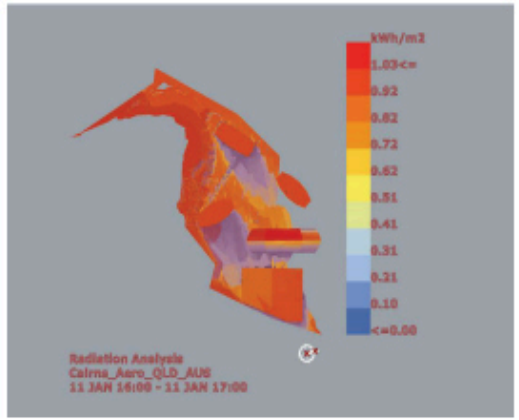
WHILE THE VISUAL REPRESENTATION AND AESTHETICS ARE A KEY FACTOR TO WHY ITERATION IS UTILISED, ADDITONALLY PARTON COMMENTED ON " DIFFERENT ITERATIONS OF THE GREATER CURVES AND HOW IT WOULD EFFECT THE TILING ANGLE" WHICH ALLUDES TO A MORE TECHNICAL PURPOSE OF ITERATION IN ORDER TO ACHIEVE DEVELOPABILITY OF A DESIGN OUTSIDE OF A DIGITAL OR ANALOG REPRESENTATION.



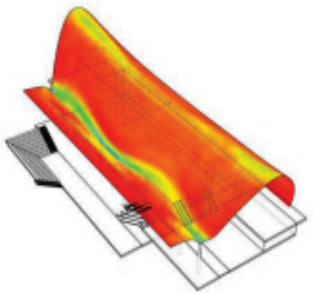


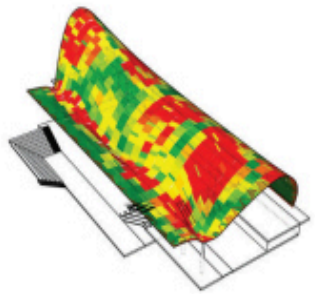


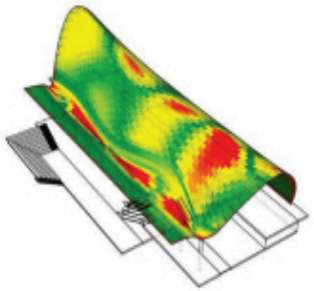


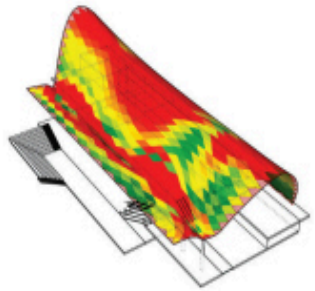


SURFACE OPTIMIZATION









REASONS FOR COMPUTATION

KEEPING WITH THE SAME EXAMPLE OF SONYA PARTON'S DISCUSSION ON THE KUTUBU CONVENTION CENTRE, THE PURPOSE OF COMPUTATION WAS ADDRESSED WITHIN THIS FRAMEWORK. PARTON TOUCHES ON GRASSHOPPER BEING UTILISED FOR THE PROCESS OF TESTING AND DESIGNING THE ROOF OF THE CENTRE, DRAWING A LINK BETWEEN ITERATION AND COMPUTATION. PARTON STATES " THERE WAS A LOT OF MODELLING AND A LOT OF REWORK... IT WAS VERY, VERY TIME CONSUMING FOR THE TEAMS... SO WE ENDED UP PUTTING IT INTO GRASSHOPER AND MAKING THE ROOF PARAMETRIC." FROM HERE PARTON CONTINUES TO DEMONSTRATE THE PURPOSE OF COMPUTATION IN TERMS OF EFFICIENCY, ANALYSIS AND OPTIMISATION.

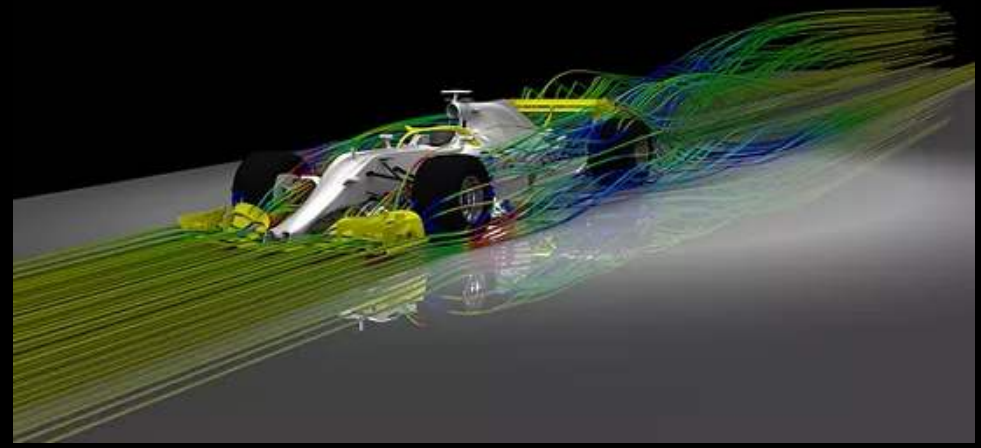
COMPUTATION ALLOWED FOR QUICK AND OPTIMISED ITERATIONS TO BE PUMPED OUT FOLLOWING THE PRODUCTION OF A PARAMETRIC BASE MODEL. COMPUTATION'S KEY PURPOSE IS TO ENABLE AND GROW UPON BASE DESIGNS, WITH THE ABILITY TO EASILY TWEAK MODELS BASED ON USER INPUT. COMPUTATION MINIMISES 'BUSY WORK' ALLOWING FOR MORE TIME AND FOCUS BEING PLACED ON THE STAKEHOLDER INPUT, MORE COMPLEX DESIGN CONCEPTS AND DEVELOPABILITY, ALL RESULTING IN THE EVENTUATION OF AN OPTIMAL END RESULT.

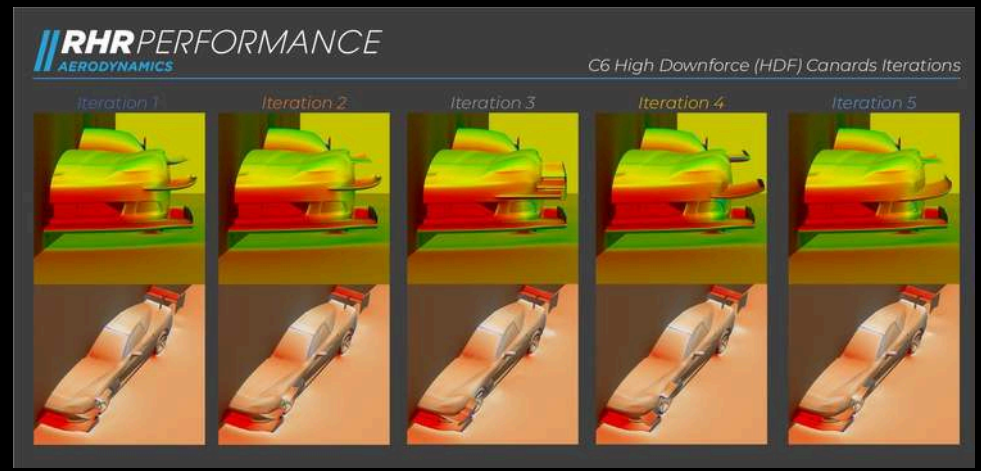
COMPUTATIONALLY SUPPORTED ITERATION

TO EXPLAIN THE VALUE OF ITERATION SUPPORTED BY COMPUTATION I WILL ADDRESS A SECTION OF MY DISCAPLINE WITH AUTOMOTIVE ENGINEERING WHICH HIGHLIGHTS ITS CRUCIAL ROLE IN DESIGN. VEHICAL AERODYNAMICS IS BUILT UPON AN ITERATIVE DESIGN MODEL, WHERE A PART IS REPEATEDLY DESIGNED, TESTED AND REMODELLED TO OPTEMISE THE AREODYNAMICS. THIS IS ITERATION. WITHOUT THIS DESIGN PHILOSOPHY A DEEPER LEARNING OF THE PART IS LOST AND ALL POSSIBLE AVENUES ARE NOT EXPLORED TO DEVELOP THE BEST FINAL RESULT.

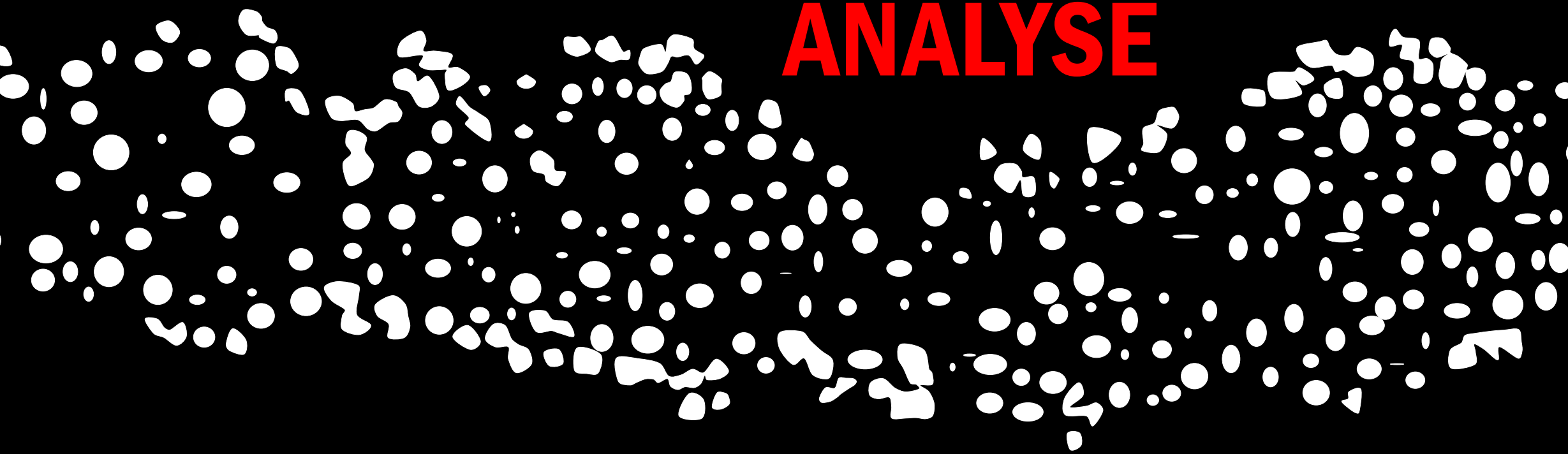
COMPUTATION IS ALSO PREVALENT IN ORDER TO SUPPORT THE ITERATION. THE AMOUNT OF DESIGNS FOR A SINGLE AERODYNAMIC CAR PART IS INCOMPREHENSIBLE. IN ORDER TO COMBAT THIS, COMPUTATIONAL DESIGN IS EMPLOYED TO MAXIMISE EFFICIENCY, AND OPTIMISE PREFORMANCE. BY SETTING PARAMETERS IN COMPUTATIONAL SOFTWARE, MANY INTERATIONS OF PARTS ARE ABLE TO BE PRODUCED EFFECTIVELY, EFFICIENTLY AND WITH A MINIMISATION OF THE LIMITATIONS OF HUMAN FUELED DESIGN.

COMPUTATIONALLY SUPPORTED ITERATION IS THE CONJUNCTION OF THE OPTIMISATION FOCUSED DESIGN PHILOSOPHY WITH THE PARAMETERISATION OF COMPUTATION. IT OVERALL SUPPORTS FAST AND EFFECTIVE DESIGN.





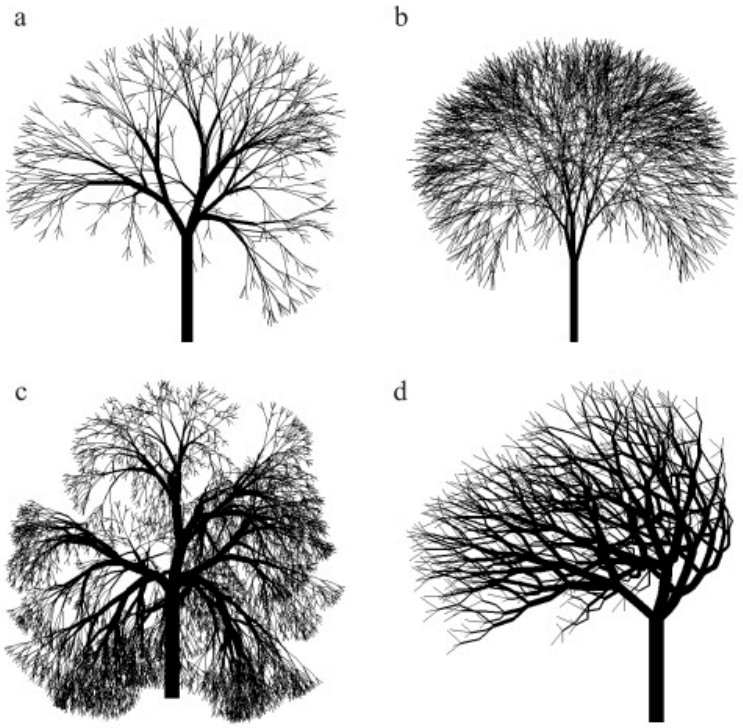
ANALYSE



ALGORITHMIC DESIGN EXAMPLE

AN EXAMPLE OF COMPUTATIONAL DESIGN THAT IS HIGHLIGHTED IN THE THE VIDEO BY DIGITAL DESIGN UNIT - TU DARMSTADT IS ALGORITHMIC DESIGN IN ARISTID LINDENMAYER AND PRZEMYSLAW PRUSINKIEWICZ'S 'THE ALGORITHMIC BEAUTY OF PLANTS'. WHERE THEY APPROACHED DEVELOPING NATURAL FORMS USING COMPUTATIONAL DESIGN. LINDENMAYER'S WORK IN THIS SPACE WAS SO IMPACTFUL IT IS STILL USED TODAY IN HOW WE MODEL TREES.

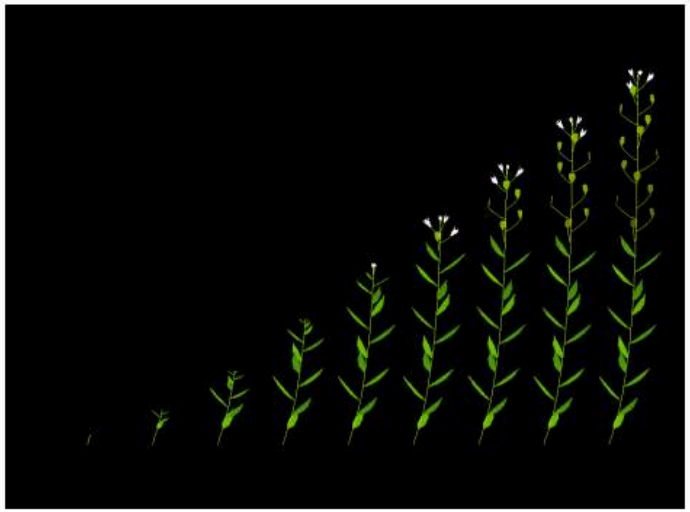
TO DEVELOP THE COMPUTATIONAL PARAMETERS REQUIRED FOR COMPUTERS TO GENERATE THESE DESIGNS, PATTERNS IN THE NATURAL FORMS OF PLANTS WAS EXPLORED. ROTATIONAL SYMMETRY OF FLOWERS, BILATERAL SYMMETRY OF LEAVES AND HELICAL ARRANGEMENTS IN PLANTS WERE SET AS PARAMETERS. THIS SET UP THE BASIS FOR L-SYSTEMS WHICH IS THE MODEL FOR PLANT COMPUTATION. HERE THE COMPUTATIONAL PROCCES COMPLETED BY COMPUTERS PROVIDED ENOUGH DETAIL FOR COMPUTER GRAPHICS TO PRESENT REALISTIC PLANT STRUCTURES.




```
#define d1 94.74        /* divergence angle 1 */
#define d2 132.63       /* divergence angle 2 */
#define a 18.95         /* branching angle */
#define lr 1.109        /* elongation rate */
#define vr 1.732        /* width increase rate */

ω  :  !(1)F(200)/(45)A
p1 :  A : *     → !(vr)F(50)[&(a)F(50)A]/(d1)
                  [&(a)F(50)A]/(d2)[&(a)F(50)A]
p2 :  F(l) : *  → F(l*lr)
p3 :  !(w) : *  → !(w*vr)
```

Figure 2.8: Examples of tree-like structures with ternary branching



```
ω   :  I(9)a(13)
p1  :  a(t) :  t>0 → [&(70)L]/(137.5)I(10)a(t-1)
p2  :  a(t) :  t=0 → [&(70)L]/(137.5)I(10)A
p3  :  A    :  *   → [&(18)u(4)FFI(10)I(5)X(5)KKKK]
                     /(137.5)I(8)A
p4  :  I(t) :  t>0 → FI(t-1)
p5  :  I(t) :  t=0 → F
p6  :  u(t) :  t>0 → &(9)u(t-1)
p7  :  u(t) :  t=0 → &(9)
p8  :  L    :  *   → [{.-FI(7)+FI(7)+FI(7)}]
                     [{.+FI(7)-FI(7)-FI(7)}]
p9  :  K    :  *   → [&{.+FI(2)--FI(2)}]
                     [&{.-FI(2)++FI(2)}]/(90)
p10 :  X(t) :  t>0 → X(t-1)
p11 :  X(t) :  t=0 → ∧(50)[[-GGGG++[GGG[++G{.].].].
                     ++GGGG.--GGG.--G.}]%
```

Figure 3.5: Development of *Capsella bursa-pastoris*. Every fourth derivation step is shown.

ITERATIVE DESIGN EXAMPLE

NOT ONLY IS ' THE ALGORITHMIC BEAUTY OF PLANTS' A PRIME EXAMPLE OF THE EARLY STAGES OF COMPUTATIONAL DESIGN, BUT IT ALSO DEMONSTRATES THE ITERATIVE PROCESS IN A FEW PERSPECTIVES.

LINDENMAYER AND PRUSINKIEWICZ'S WORK REVIEWS INTERATION WITHIN THE COMPUTATIONAL PROCESS TO BE AN ADDITIVE STEP TO EACH LAYER OF THE MODELLING OF THE NATURAL OBJECT, WHERE THE PARAMETERS ITERATIVELY INTRODUCE MORE ASPECTS OF THE NATURAL FORM.

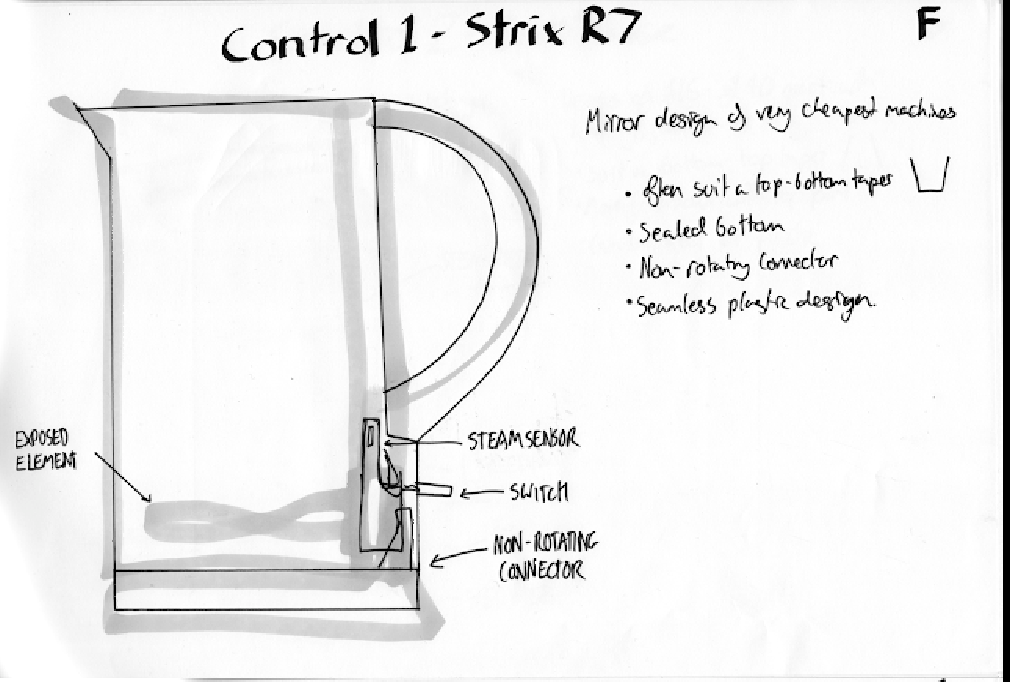
THEIR WORK ALSO ADDRESSES ITERATION IN DESIGN AS THE REVIEW, FEEDBACK, EDIT AND PROCEED PROCESS THAT IS CRUCIAL TO REFINING AND DEVELOPING DESIGNS.

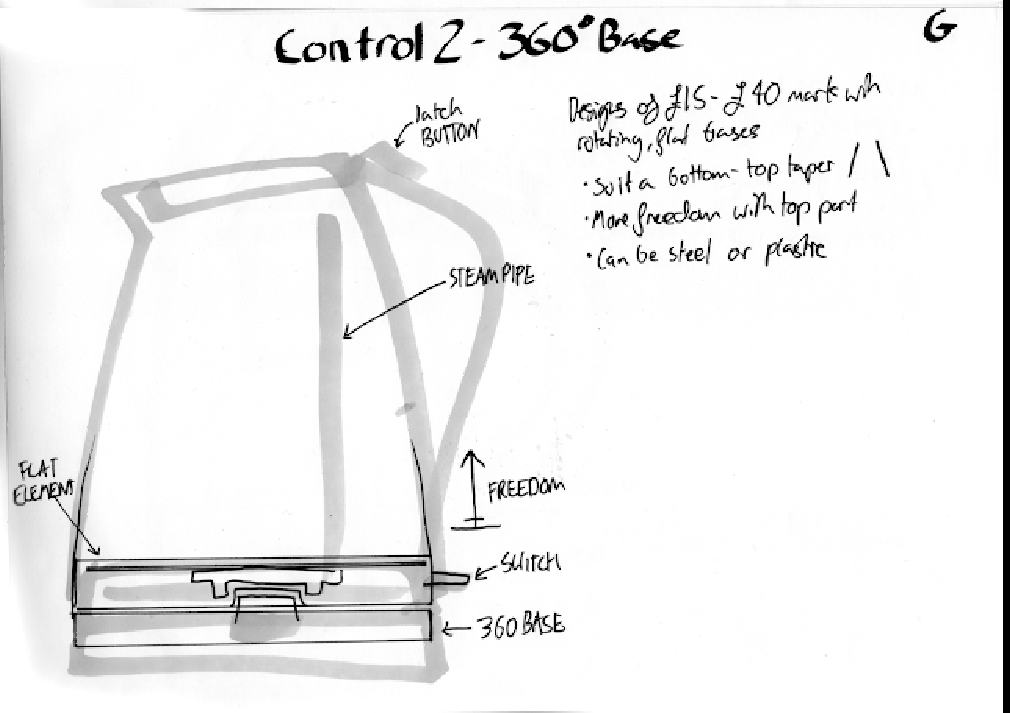
COMPUTATIONAL DESIGN EXAMPLE

AN EXAMPLE OF COMPUTATIONAL DESIGN OUTSIDE MY DISCIPLINE IS WITH FRANCIS BITONTI'S VERLAN DRESS. WHILE THIS DRESS DOES EMPLOY FUTURE FOCUSED TECHNIQUES OF MANUFACTURING, THE USE OF COMPUTATIONAL DESIGN IS WITHIN " PARAMETERS IN ORDER TO CUSTOMISE THE DRESS". BITONTI IN COLABORATION WITH 3D MODELLING COMPANY THINGYVERSE PROGRAMMED A COMPUTATIONAL SYSTEM THAT TAKES IN SIZE PARAMETERS FROM A USER AND PRODUCES A 3D PRINTABLE VERSION OF THE VERLAN DRESS.

THIS COMPUTATIONAL DESIGN SYSTEM ALTERS THE PATTERN OF THE DRESS, THE SHAPE AND THE FORM BASED ON THE PARAMETERS SET BY THE USER. THIS CONCEPT PUSHED THE BOUNDS OF TRADITIONAL COMPUTATIONAL DESIGN THROUGH A MORE WIDESPREAD INPUT NETWORK IN COMPARISON TO THE TYPICAL COMMERCIALISED USE OF COMPUTATIONAL DESIGN.







ITERATIVE DESIGN EXAMPLE

ITERATIVE DESIGN IS ESSENTIAL FOR THE PRODUCT DESIGN PROCESS. NOT ONLY DOES IT ALLOW FOR THE PRODUCTION OF A WELL ANALYISED AND QUALITY END RESULT, BUT ALSO ALLOWS FOR COLABORATIVE IMPACT ON A FUTURE ITERATION.

TAKE THE EXAMPLE OF TOM LEVER'S KETTLE ITERATIONS ON HIS BLOG WHERE HE POINTS OUT THAT THE CHOOSEN DESIGN IS AN AMALGAMATION OF 6 THEMES, 5 IDEAS, 3 CONTROLS WITH A TOTAL OF 48 CONCEPTS.

ITERATION IS UTILISED IN DESIGN BY LARGE SCALE COMPANIES AIMING TO COMMUNICATE TO STAKEHOLDERS, AND BY FREELANCE DESIGNERS. THIS IS BECAUASE IT IS QUINTESENTIAL TO HIGH QUALITY PURPOSE BUILT DESIGNS.

COMPUTATION

PROS

- ALLOWS FOR A PRECISE AND EFFICIENT DESIGN PROCESS, SAVING TIME AND RESOURCES
- ITS LINK TO ITERATION ALLOWS FOR RAPID PROTOTYPING AND THE ABILITY TO MAKE TINY CHANGES THOUGH THE PARAMETRIC MODELS
- THE FLEXABILITY OF COMPUTATIONAL DESIGN ALLOWS IT TO BE APPLIED TO MANY DISCAPLINES

CONS

- EXPENSIVE TO IMPLEMENT ON A LARGE SCALE AND REQUIRES SPECIALISED TRAINING
- PHILISOPHICAL DEBATE ON THE REMOVAL OF CREATIVITY DUE TO AN OVERRELIANCE ON COMPUTATIONAL DESIGN

ITERATION

PROS

- OPTIMISED END RESULT, FORMULATED AS AN EMALGIMATION OF ALL THE BEST FEATURES OF MANY ITERATIONS
- ALLOWS FOR CONTINUAL IMPROVEMENTS BASED ON A DIALODGE BETWEEN DESIGNERS AND STAKEHOLDERS
- PRODUCES MORE TIME TO DESIGN AND IMPROVE THOUGH COMMUNICATING VISUALLY WITH RAPID PROTOTYPING
- GENERATES A DESIGN PATH TO RETRACE AND EXPLORE PAST ITERATIONS TO BETTER UNDERSTAND THE FINAL PRODUCT

CONS

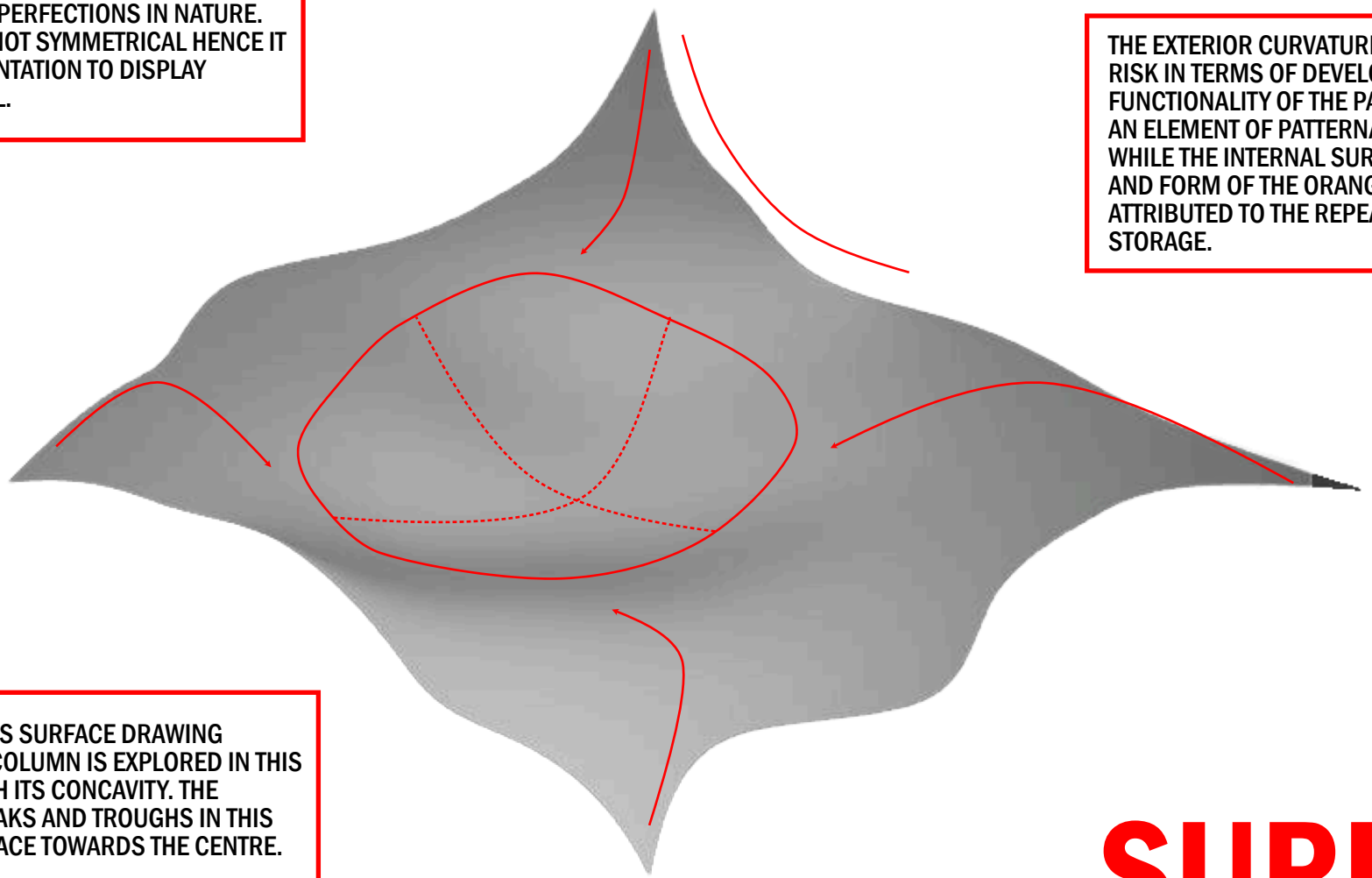
- CAN POTENTIALLY GENERATE WASTED TIME ON EXPLORING NEW ITERATIONS WHEN A PREVIOUS DESIGN IS OPTIMAL
- REQUIRES A LOT OF PROJECT MANAGEMENT AND RESOURCES



PRACTISE

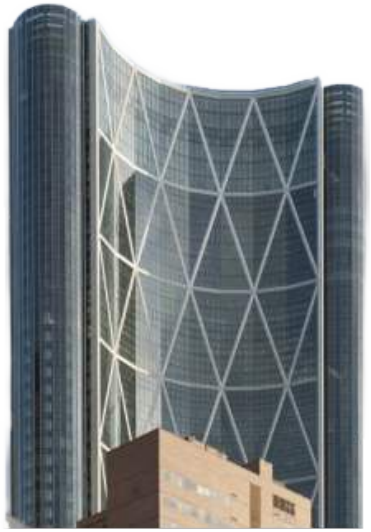
THE NON UNIFORM TROUGH IN THE CENTER OF THE SURFACE MIRRORS THE INPERFECTIONS IN NATURE. THE ORANGE SURFACE IS NOT SYMMETRICAL HENCE IT WOULD BE A MISREPRESENTATION TO DISPLAY UNIFORMITY IN THE MODEL.

THE EXTERIOR CURVATURE OF THE SURFACE, WHILE A RISK IN TERMS OF DEVELOPABILITY AND FUNCTIONALITY OF THE PANEL EXTRUSIONS, BRINGS AN ELEMENT OF PATTERNATION TO THE SURFACE. WHILE THE INTERNAL SURFACE PLAYS ON THE TEXTURE AND FORM OF THE ORANGE, THE EXTERNAL SHAPE IS ATTRIBUTED TO THE REPEATED PATTERN OF THE JUICE STORAGE.



TENSION IN THE ORANGE'S SURFACE DRAWING TOWARDS THE CENTRAL COLUMN IS EXPLORED IN THIS SURFCE MODEL THROUGH ITS CONCAVITY. THE CONJUNCTION OF THE PEAKS AND TROUGHS IN THIS SURFACE PULL THE SURFACE TOWARDS THE CENTRE.

SURFACE



ARTIFICIAL SYSTEMS
THE BOW BY FOSTER AND PARTNERS

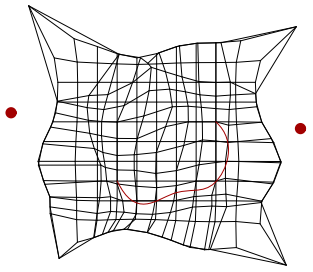
BIOTIC
RED BLOOD CELL





ABIOTIC
WATER MINISCUS

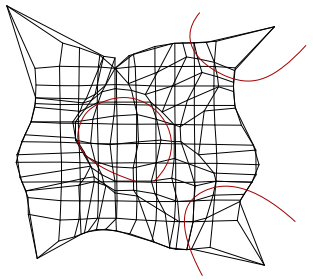
SURFACE



G01

Top:
ptOffsetPoints
attractor points
min: 70
max: 100
away

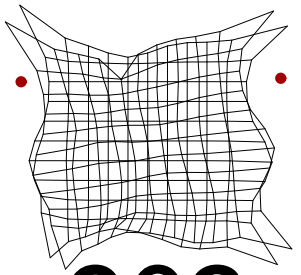
ptShuffleGrid
attractor curve
magnitude 3



G02

Top:
ptOffsetPoints
attractor curves
min: 100
max: 120
away

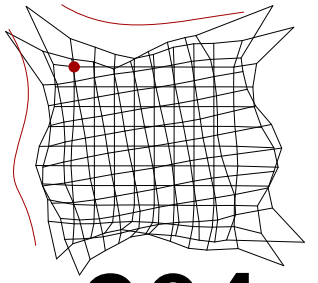
ptShuffleGrid
attractor curve
magnitude 3



G03

Top:
ptOffsetPoints
attractor points
min: 70
max: 120
away

rotate 10 degrees



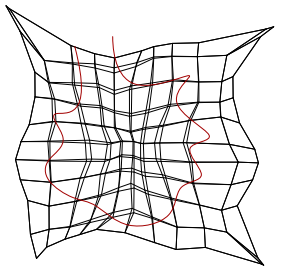
G04

Top:
ptshufflegrid
attractor points
magnitude 2:
towards

rotate 10 degrees

Bottom:
ptShuffleGrid
attractor curves
away
magnitude 1.5

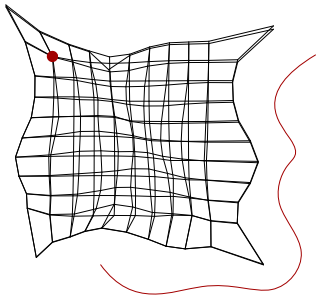
ptOffsetPoints
point attractors
min: 40
max: 120
towards



G05

Top:
ptOffsetPoints
attractor curves
min: 70
max: 120
away

Bottom:
ptshufflegrid
curve attractor
away
magnitude 1.4

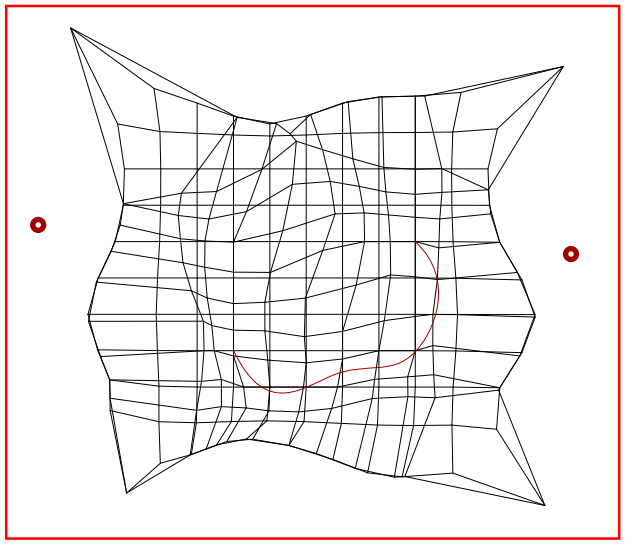


G06

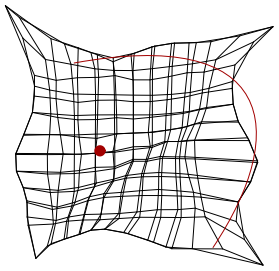
Top:
ptoffsetpoints
attractor point
min: 60
max: 120
towards

z rotation 10

Bottom:
ptshufflegrid
curve attractor
away
magnitude 1.2



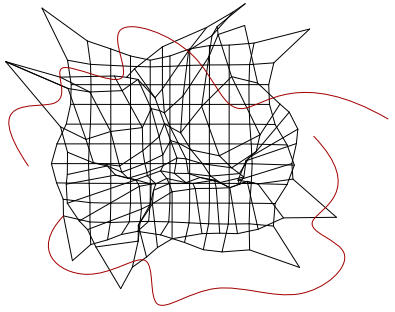
GRIDS



G07

Top:
ptshufflepoints
attractor points
towards
magnitude 1.2

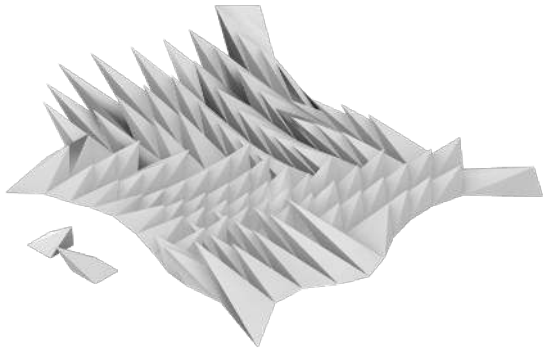
Bottom:
ptshufflegrid
curve attractor
away
magnitude 1.8



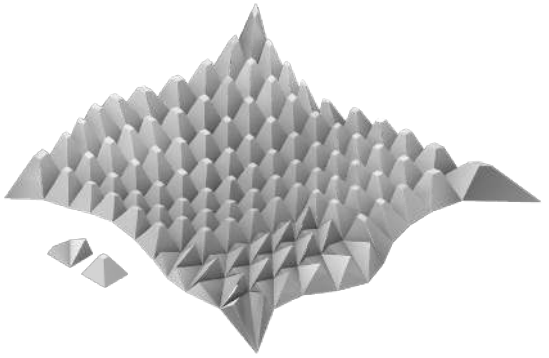
G08

Top:
ptshuffle grid
attractor curves
away
magnitude 2

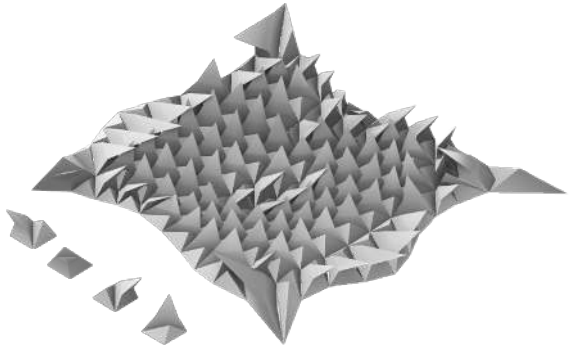
rotate 20 degrees



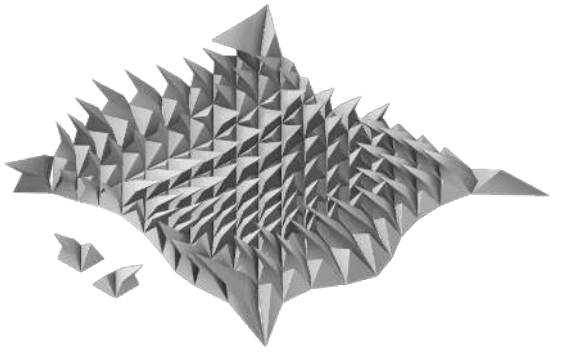
P01



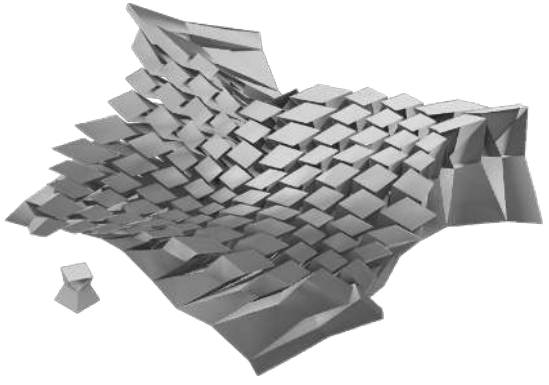
P02



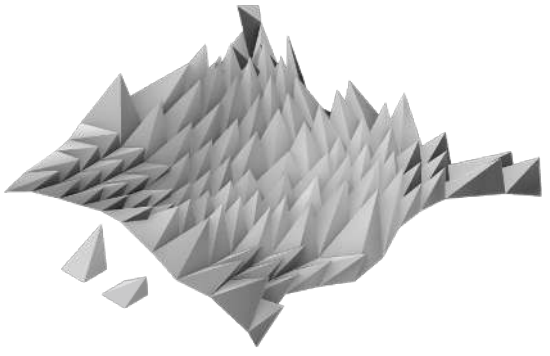
P03



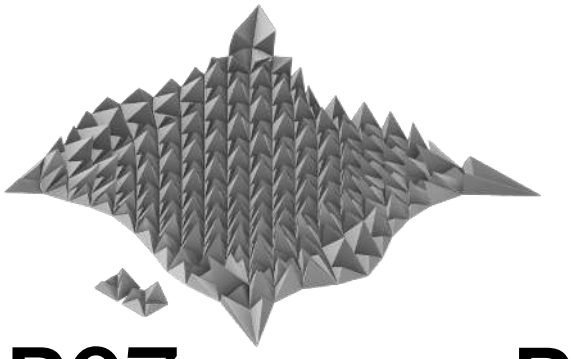
P04



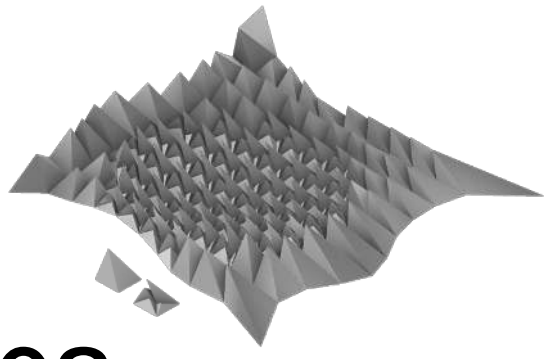
P05



P06

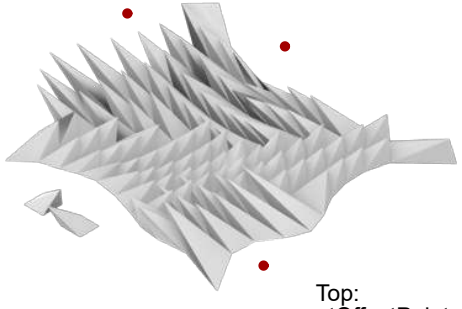


P07



P08

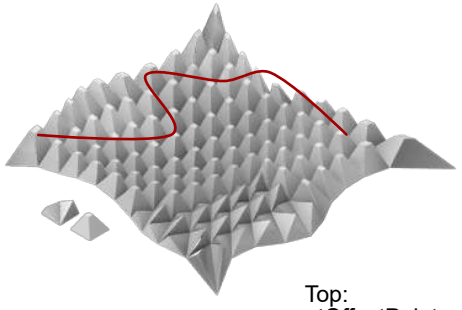
PANELS



P01

Top:
ptOffsetPoints
attractor points
away
magnitude 1.8

2 panel objects

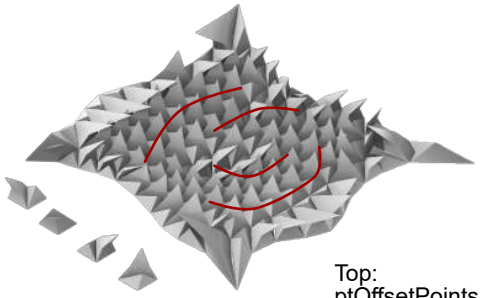


P02

Top:
ptOffsetPoints

ptShuffleGrid
attractor curve
towards
magnitude 1

2 panel objects

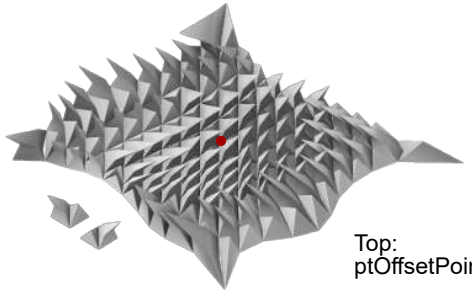


P03

Top:
ptOffsetPoints

ptShuffleGrid
attractor curve
towards
magnitude 1

4 panel objects

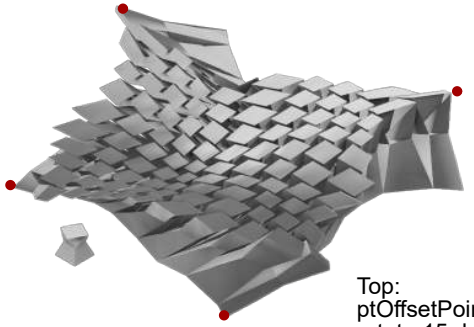


P04

Top:
ptOffsetPoints

ptShuffleGrid
attractor point
towards
magnitude 1.2

4 panel objects

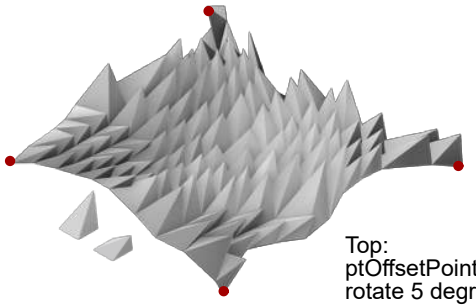


P05

Top:
ptOffsetPoints
rotate 15 degrees

Bottom:
ptShufflegrid
attractor points
towards
magnitude 1.6

1 panel object

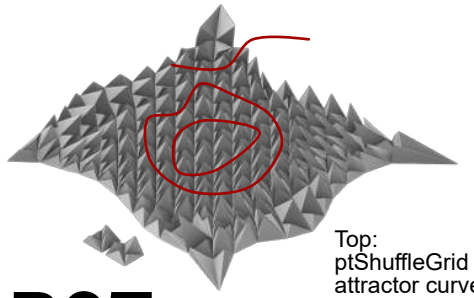


P06

Top:
ptOffsetPoints
rotate 5 degrees

Bottom:
ptShufflegrid
attractor points
towards
magnitude 1.6

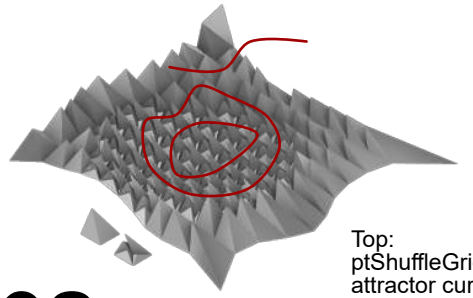
2 panel objects



P07

Top:
ptShuffleGrid
attractor curves
towards
magnitude 1.4

2 panel objects

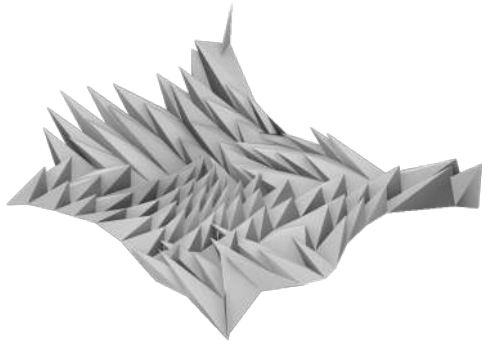


P08

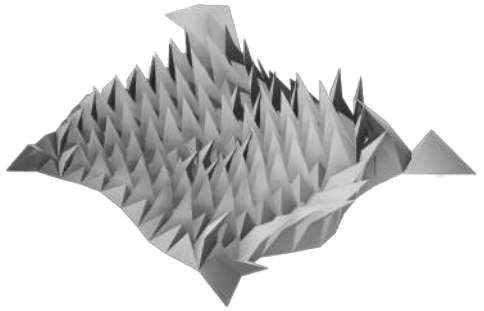
Top:
ptShuffleGrid
attractor curves
towards
magnitude 1.4

2 panel objects

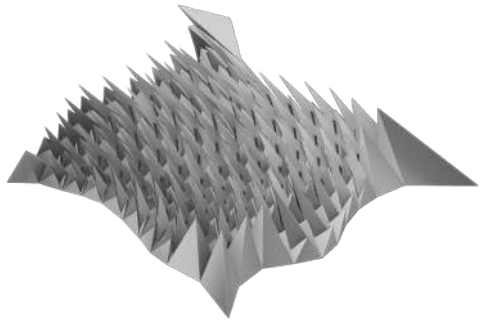
PANEL ANALYSIS



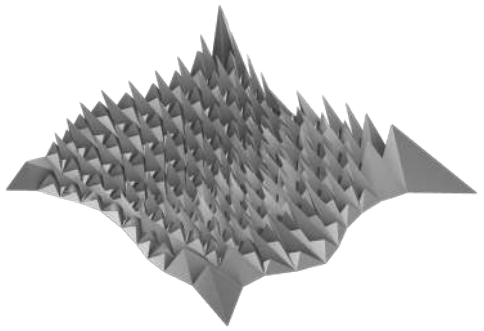
H01
G05 + P01



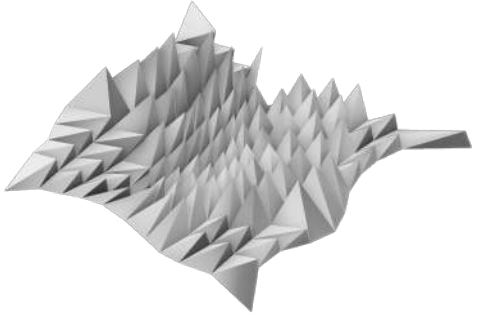
H02
G04 + P03



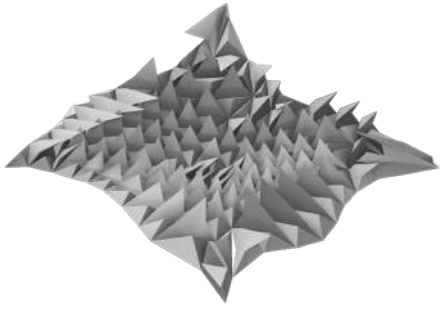
H03
G04 + P08



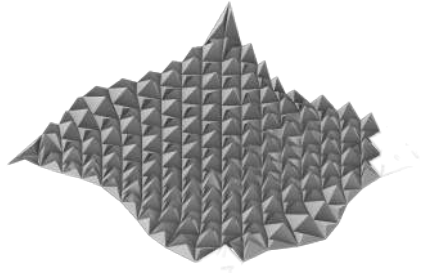
H04
G06 + P08



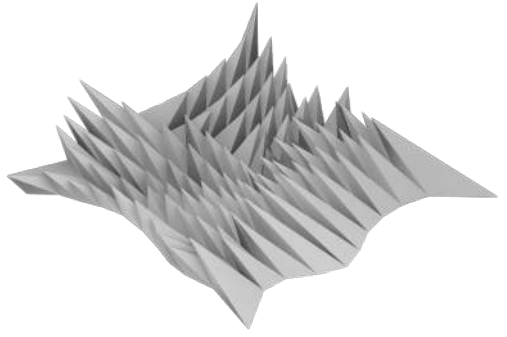
H05
G03 + P06



H06
G07 + P03



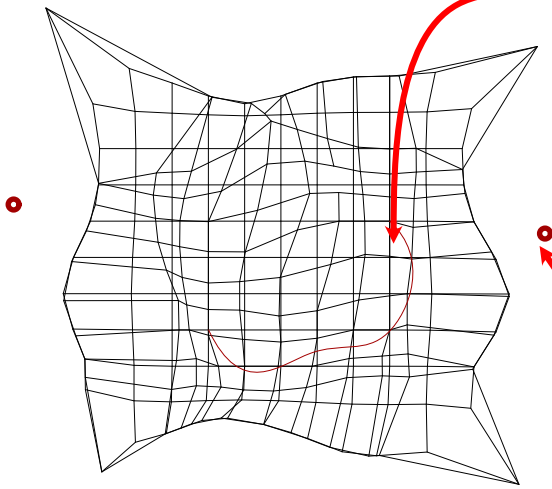
H07
G02 + P07



G06 + P01

HYBRIDS

G01

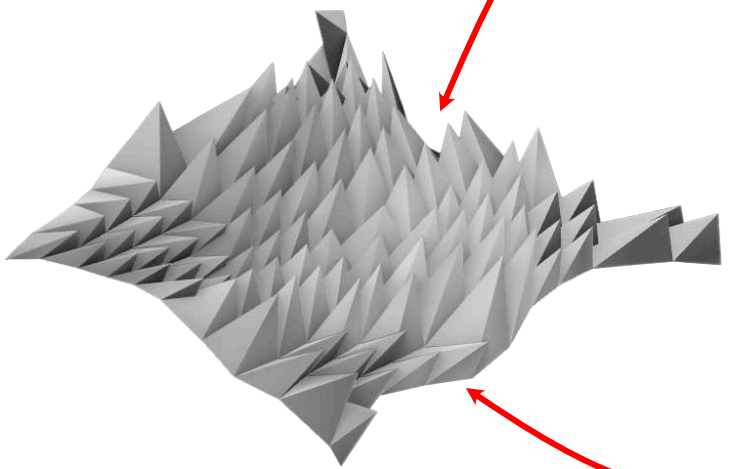


THE ATTRACTOR CURVE IN THE CENTER OF THE GRID PROVIDES THE BEST SIMULATION OF THE TENSION DRAWN TOWARDS THE CENTRAL COLUMN OF THE ORANGE. ITS IRREGULAR SHAPE COMMENTS ON THE NON UNIFORMITY OF THE ORANGE.

THE TWO ATTRACTOR POINTS FURTHER EMPHASISE THE DRAW TOWARDS THE MIDDLE WHILE ALSO PUSHING THE GRID TO EMULATE THE WAVE-LIKE PATTERNATION OF THE JUICE STORAGE.

+

P06

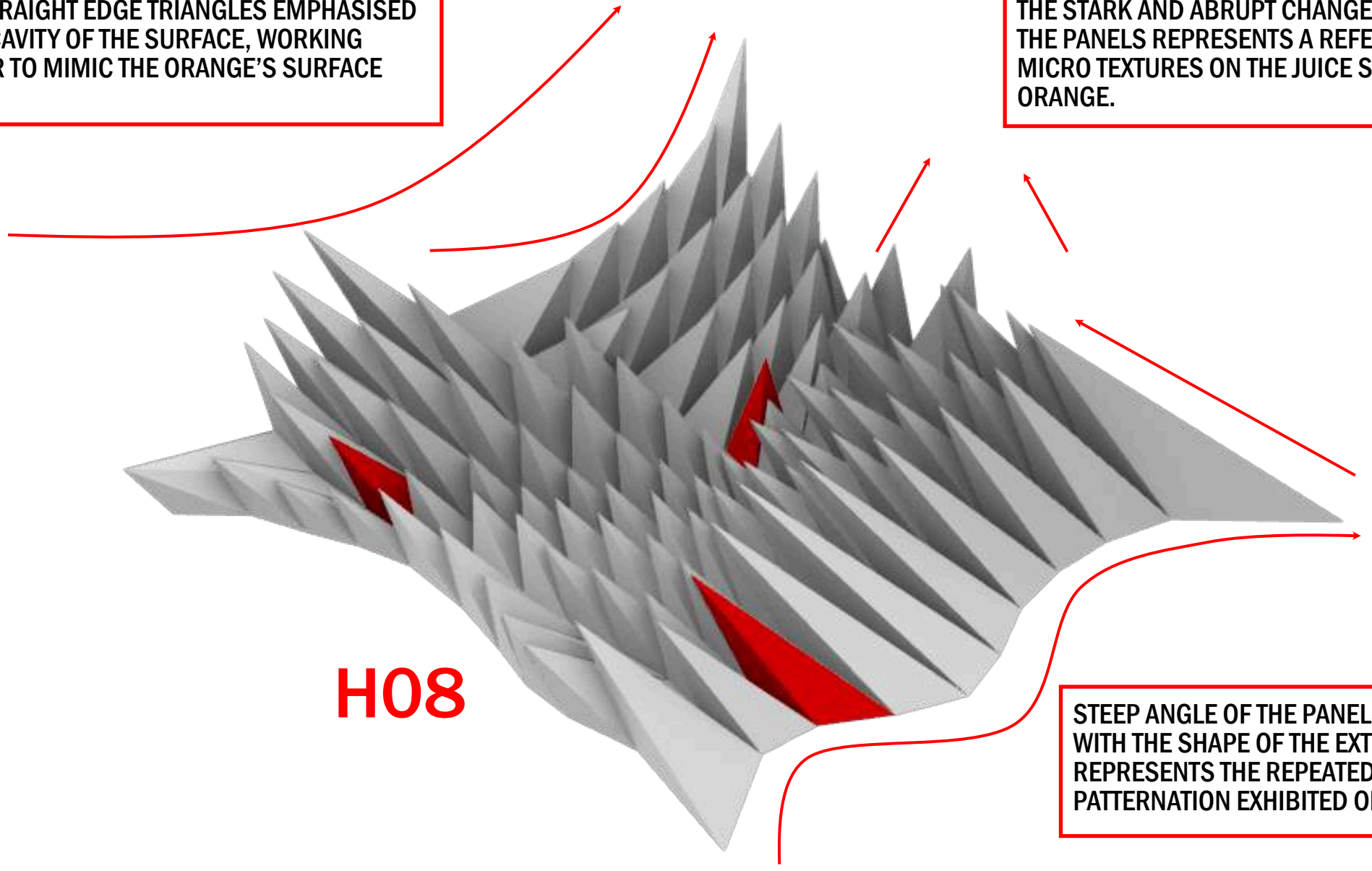


WHILE SIMPLISTIC IN APPEARANCE, THESE PANELS BRING COMPLEXITY IN THE EFFECTS THEY HAVE DUE TO BEING PULLED OUTSIDE THEIR BOUNDING BOXES. WHILE ALL THE TRIANGLES HAVE STRAIGHT EDGES, THEIR ANGLE AND MOVEMENT WITH EACH OTHER GENERATES A CURVED APPEARENCE, PUSHING THE NARRATIVE OF THE ORANGE'S CURVED PATTERNATION.

EVENTHOUGH OTHER PANELS EXPLORED IN THE PANEL STUDY ARE MORE VISUALLY COMPLEX AND EXPLORE DIFFERENT FACITS OF THE ORANGES SURFACE, THEY LACKED THESE PANEL'S LEVEL OF DEVEOLPABILITY. AS SOMEONE WITHIN THE DISCIPLINE OF ENGINEERING, FABRICATION IS A CORE FACTOR I SEE BEING CRUCIAL TO MY FUTURE DESIGNS. BECASUSE OF THIS I LEANED TOWARDS A MORE DEVELOPABLE PANEL RATHER THAN CHOOSING AN EXTREME DESIGN.

CURVATURE CAUSED BY LAYERING AND ANGLES OF THE STRAIGHT EDGE TRIANGLES EMPHASISED THE CONCAVITY OF THE SURFACE, WORKING TOGETHER TO MIMIC THE ORANGE'S SURFACE FORM.

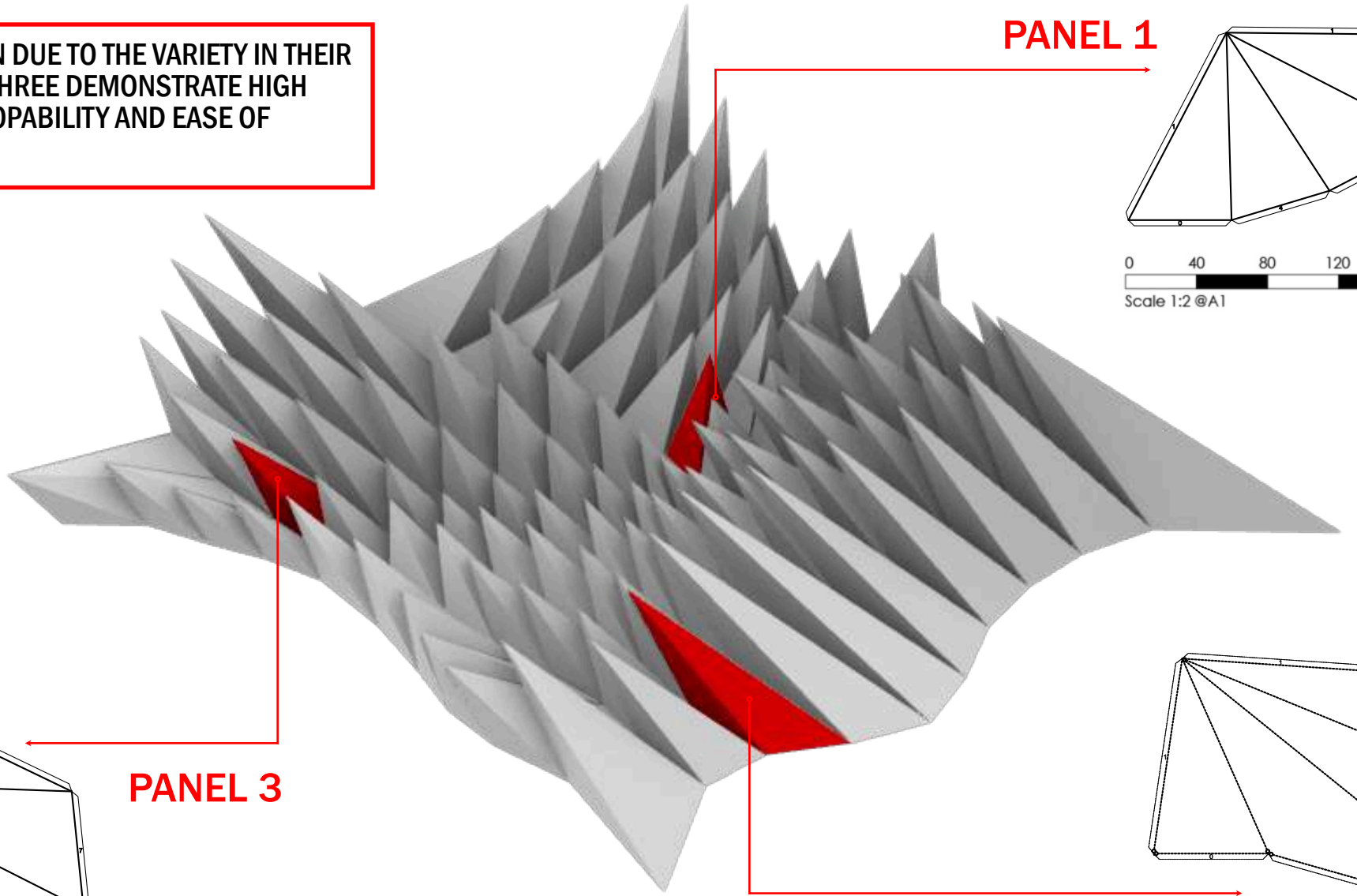
THE STARK AND ABRUPT CHANGE IN ANGLE OF THE PANELS REPRESENTS A REFERENCE TO THE MICRO TEXTURES ON THE JUICE STORAGE OF THE ORANGE.



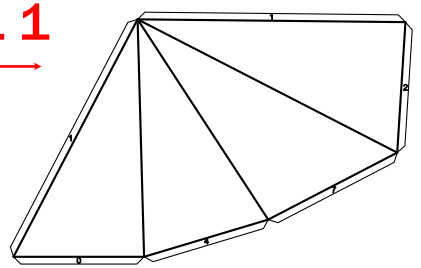
H08

STEEP ANGLE OF THE PANELS IN CONJUCTION WITH THE SHAPE OF THE EXTERNAL SURFACE REPRESENTS THE REPEATED BUT ASYMMETRICAL PATTERNATION EXHIBITED ON THE ORANGE.

3 PANELS CHOSEN DUE TO THE VARIETY IN THEIR STEEPNESS, ALL THREE DEMONSTRATE HIGH LEVELS OF DEVELOPABILITY AND EASE OF FABRICATION.

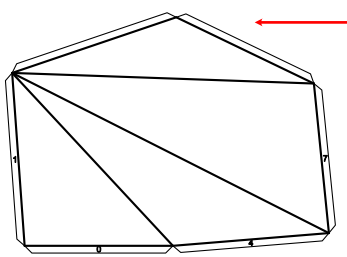


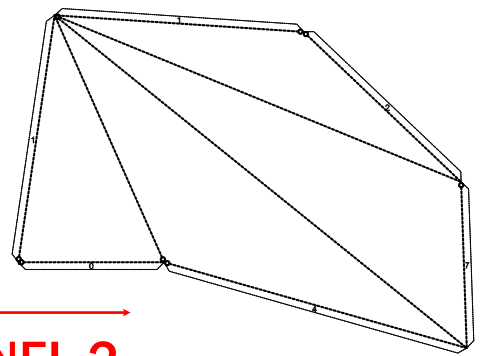
PANEL 1





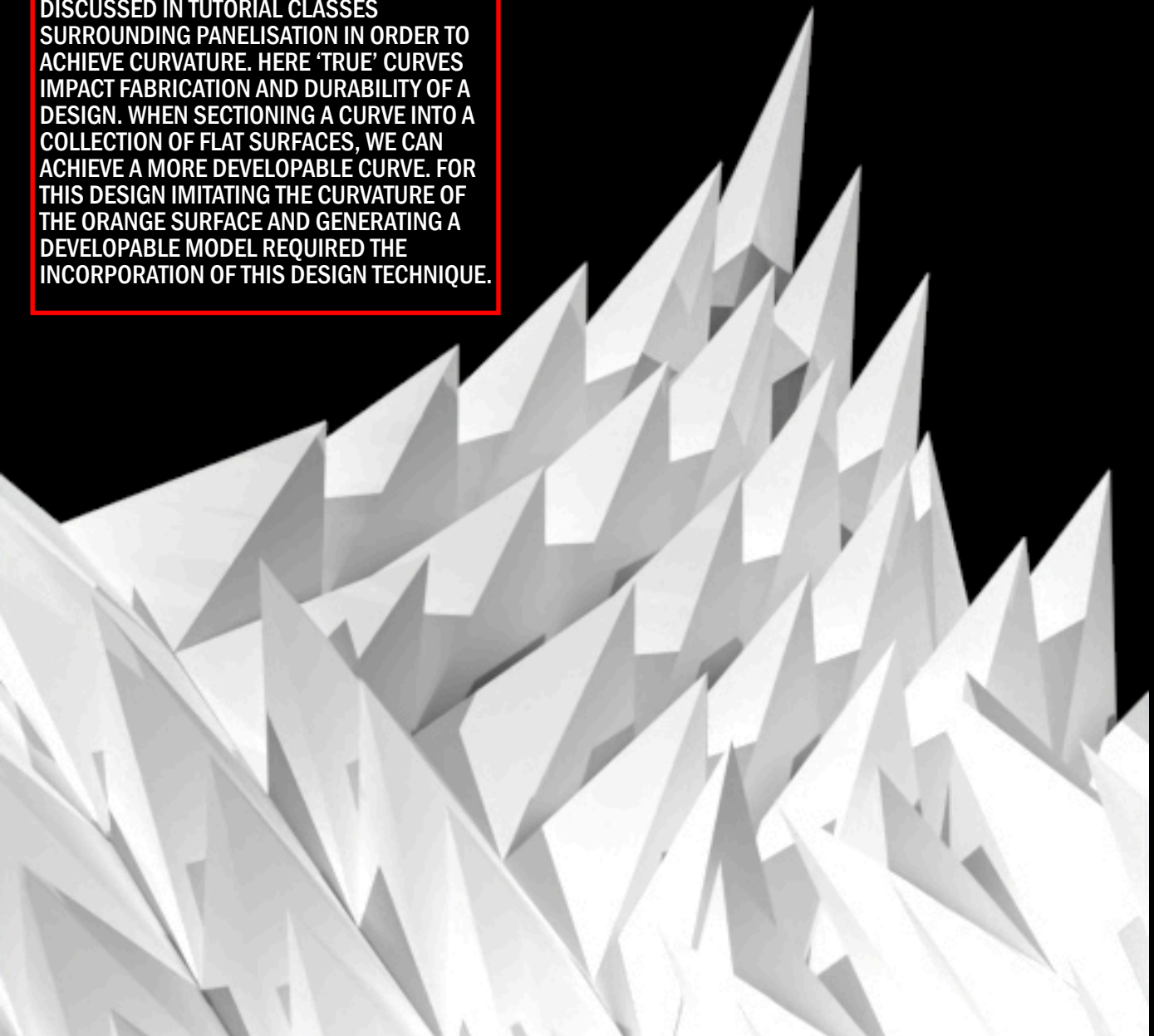
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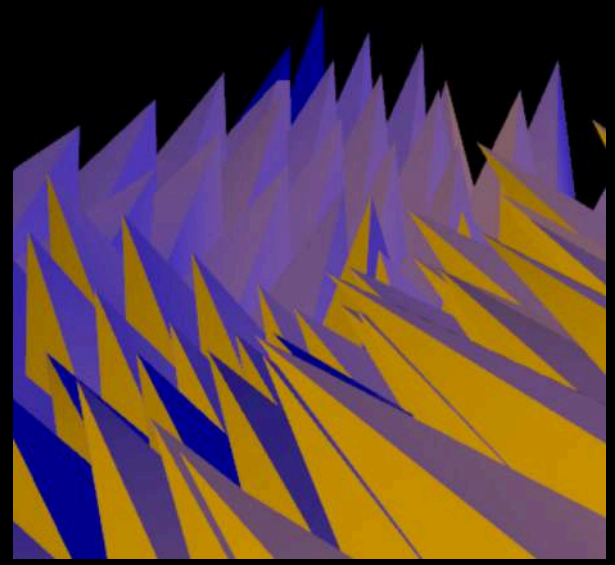


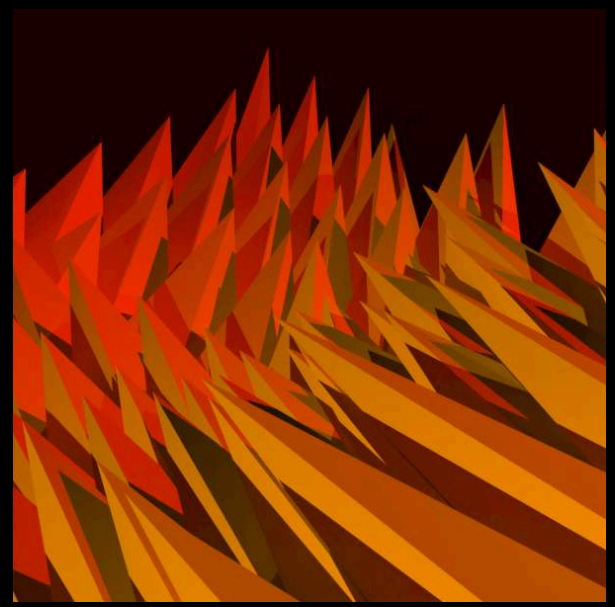


PANEL 2

THESE ZOOMED IN RENDERS OF THE SURFACE OF THE MODEL DRAW ATTENTSION TO A TOPIC DISCUSSED IN TUTORIAL CLASSES SURROUNDING PANELISATION IN ORDER TO ACHIEVE CURVATURE. HERE ‘TRUE’ CURVES IMPACT FABRICATION AND DURABILITY OF A DESIGN. WHEN SECTIONING A CURVE INTO A COLLECTION OF FLAT SURFACES, WE CAN ACHIEVE A MORE DEVELOPABLE CURVE. FOR THIS DESIGN IMITATING THE CURVATURE OF THE ORANGE SURFACE AND GENERATING A DEVELOPABLE MODEL REQUIRED THE INCORPORATION OF THIS DESIGN TECHNIQUE.

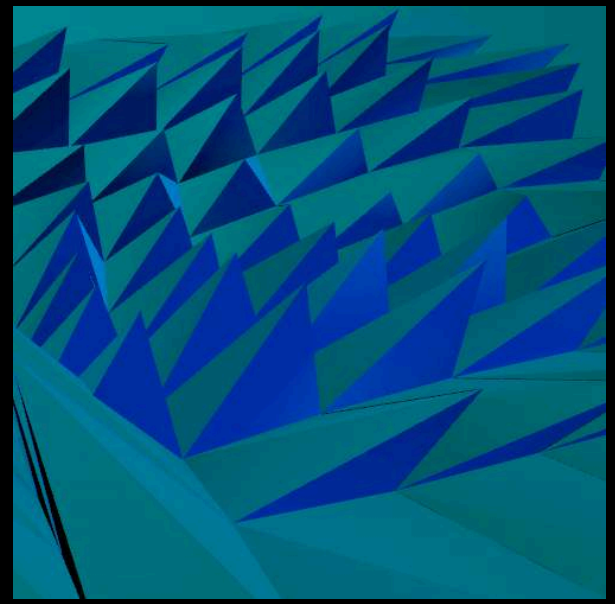


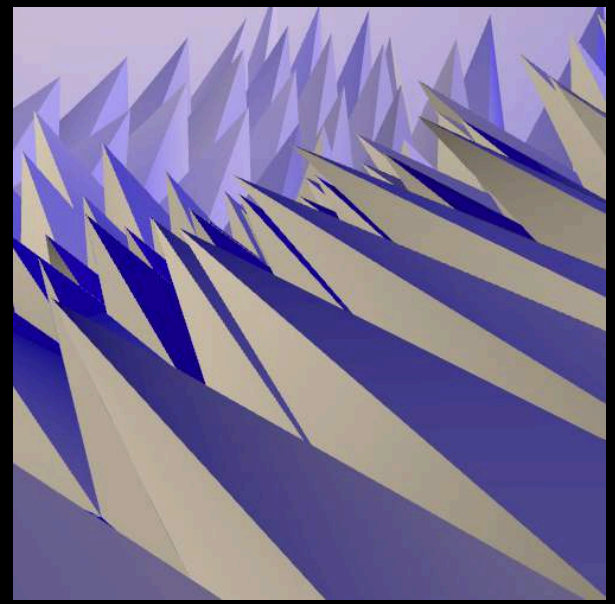




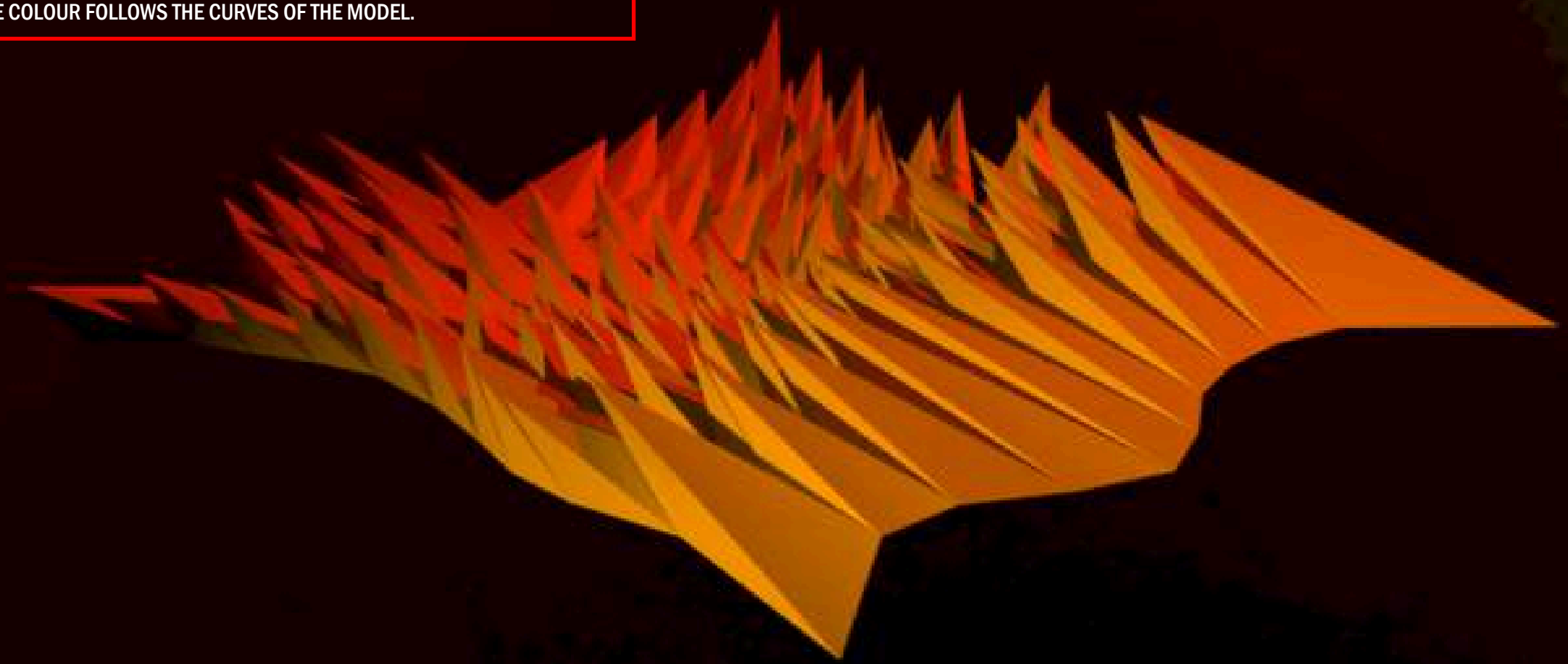
TO HIGHLIGHT KEY FEATURES OF THE RENDERS, LIGHTING HAS BEEN IMPLEMENTED. THE OPPOSING ANGLES OF THE PANELS BEING A KEY FEATURE THAT REPRESENTS THE TENSION OF THE ORANGE SURFACE BEING PULLED TOWARDS THE CENTRE, WHICH IS EVIDENT WITH LARGE SHADOWS BEING CAST UPON THE CENTER. ANOTHER KEY FEATURE IS THE GRADUATION OF THE ANGLES OF THE PANELS DEVELOPING A PATTERNATION PROMPTED BY BOTH THE ATTRACTOR CURVE AND THE ATTRACTOR POINTS. THIS PATTERNATION IS REPRESENTATIVE OF THE TEXTURE OF THE JUICE STORAGE OF THE ORANGE.







IN ORDER TO LINK BACK TO THE EXAMPLE OF COMPUTATIONALLY SUPPORTED ITERATION THAT I EXPLORED IN THE UNDERSTAND SECTION IN AN AETHETIC MANNER, I EXPLORED COLOURED LIGHTING IN RENDERS. THE CFD RENDERS IN AUTOMOTIVE ENGINEERING HAVE VERY SPECIFIC HUES OF COLOUR FOR THE PURPOSE OF VISIBILITY AND IDENTIFICATION OF DISPARITY. I TOOK INSPIRATION FROM THE COLOURS AND FLOW OF THE CFD RENDERS AND EMULATED IT IN THIS RENDER WITH NOT ONLY THE COLOUR BUT ALSO WITH THE WAY THE COLOUR FOLLOWS THE CURVES OF THE MODEL.

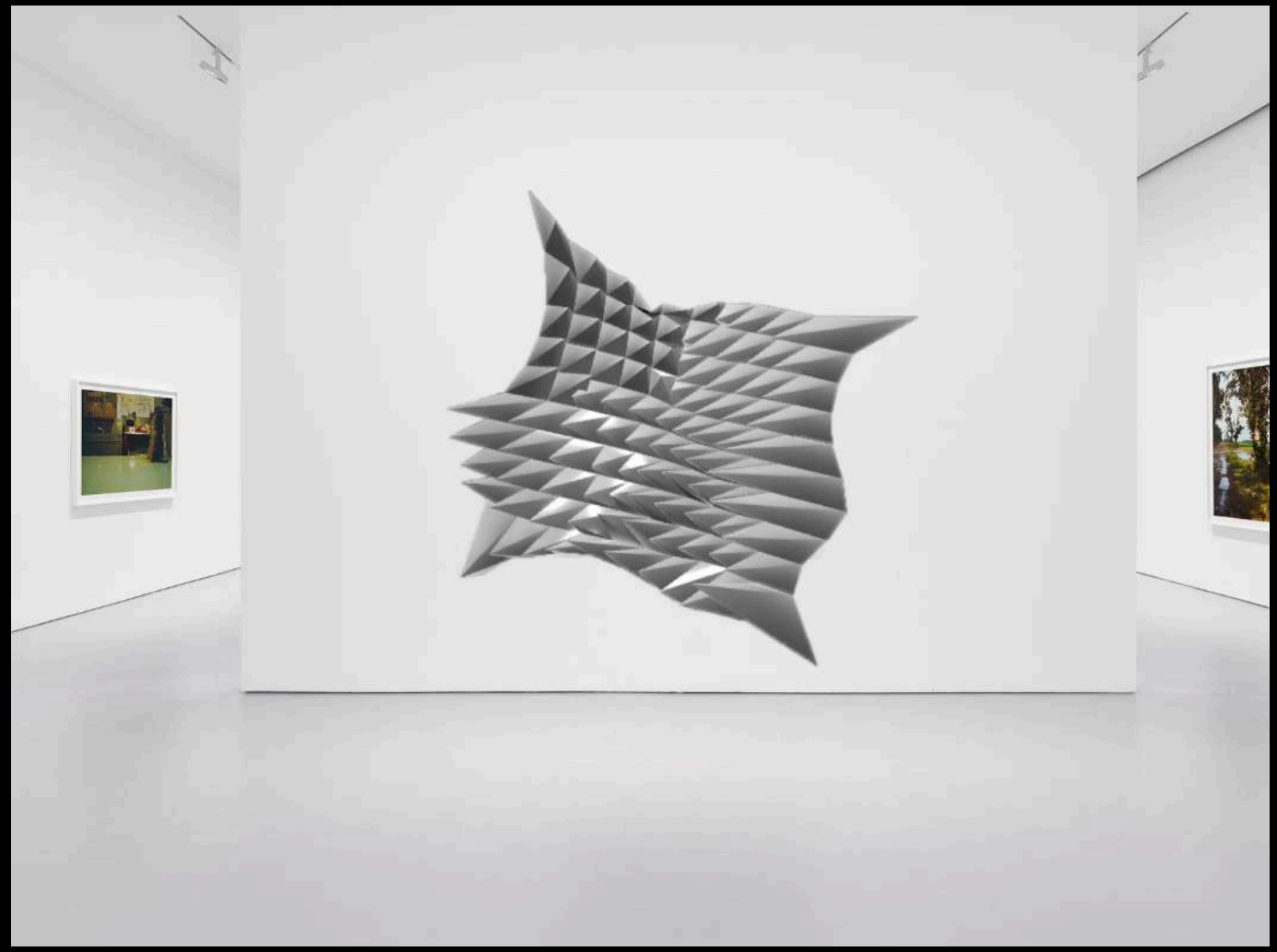


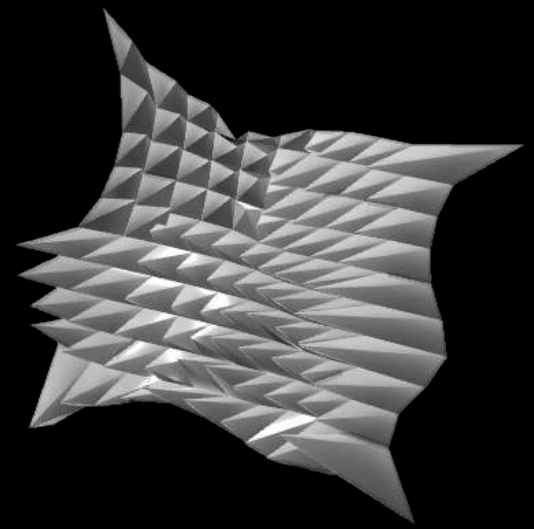
PHYSICAL REALISATION

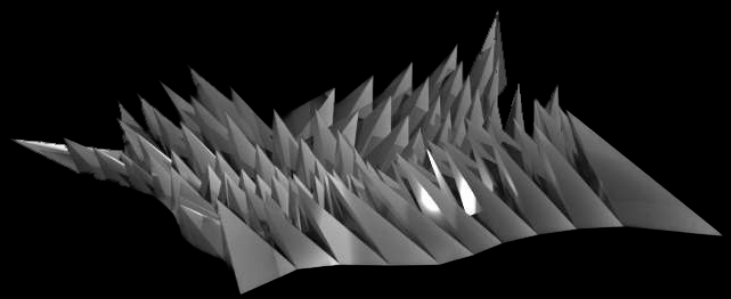
TO EXTEND THE PERCEPTION OF THE MODEL AND POTENTIAL SPACES IT COULD EXIST IN, I HAVE PRESENTED IT IN SOME DIFFERING FORMS. THIS INITAL ONE FOCUSES ON SCALE THROUGH THE USE OF THE MODEL AS AN ART INSTALLATION.

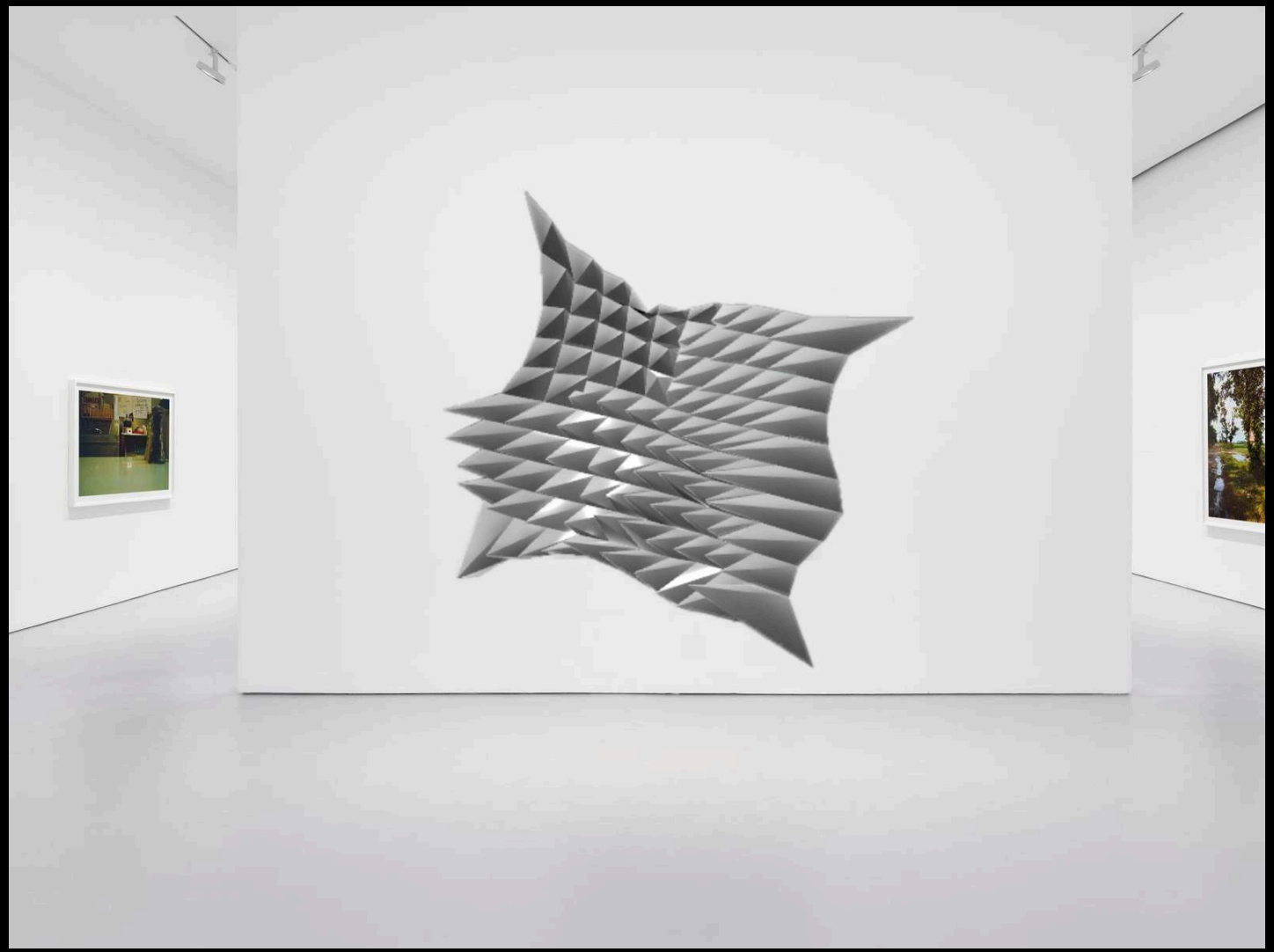
CONCEPTUALISING THE SURFACE AS A PHYSICAL MODEL AIDED IN THE PATH TO ACHIEVE DEVELOPABILITY, ESSPECIALLY AS SCALING INTRODUCES COMPLICATIONS WITH THE WEIGHT AND STRUCTURAL INTEGRITY OF THE MODEL.

ASIDE FROM THE DEVELOPABILTY, OTHER ASPECTS THAT THE ART INSTALLATION VERSION HIGHLIGHTS IS TEXTURE AND LIGHTING. BY HAVING THE TRIANGLES ACT AS A PROTRUSIONS OUT FROM THE WALL, THE WAY THE LIGHT BOUNCES OFF EACH PANEL IS AN IMPORTANT ASPECT OF THIS DESIGN.









PHYSICAL REALISATION

USING MY LEARNINGS FROM FRANCIS BITONTI'S VERLAN DRESS SURROUNDING FASHION AND COMPUTATION, WHILST PUSHING IT TOWARDS A MORE CREATIVE UTELISITION OF COMPUTATION RATHER THAN FOR FUNCTIONALITY, I HAVE IMAGINED MY MODEL AS BOTH A TEXTURED SKIRT AND JEWELLERY.

THROUGH THIS PROCESS THE EVENTUATION OF COMPUTATIONAL DESIGN CAN BE SEEN THROUGH AN AESTHETIC DRIVEN LENS WHERE OPTIMISATION IS FRAMED IN PERFECTING THE MODEL'S BEAUTY ALONGSIDE ITS FUNCTIONALITY.

THESE MOCK UPS OF FASHION PIECES OF MY MODEL EMPHASISE ITS KEY FEATURES OF TEXTURE, SHAPE, SURFACE AND PATTERNATION WHILE BRINGING THEM TO A NEW LEVEL. THE BOLD JEWELLERY HIGHLIGHTS TEXTURE AND THE MODEL'S CONCAVITY MORE SO THAN THE SKIRT, WHICH PLACES EMPHASIS ON PATTERNATION AND LIGHTING AS WE CAN IMAGINE THE MOVEMENT OF THE SKIRT WOULD MAKE THE LIGHTING EFFECTS ON THE PANELS FLUID RATHER THAN RIGID.

THE WORLD OF DESIGN IS EVER MOVING FORWARD, ADAPTING TO NEW DEVELOPMENTS AND EMBRACING OPTIMISATION. WITHIN THIS I BELIEVE IT IS EASY TO FORGET THAT THESE OPTIMISATIONS, FOR EXAMPLE COMPUTATIONAL DESIGN, ARE NOT ONLY FOR EFFICIENCY, PRECISION AND FLEXABILITY, THEY ARE ALSO FOR AESTHETICS AND APPEAL.






IMAGINE

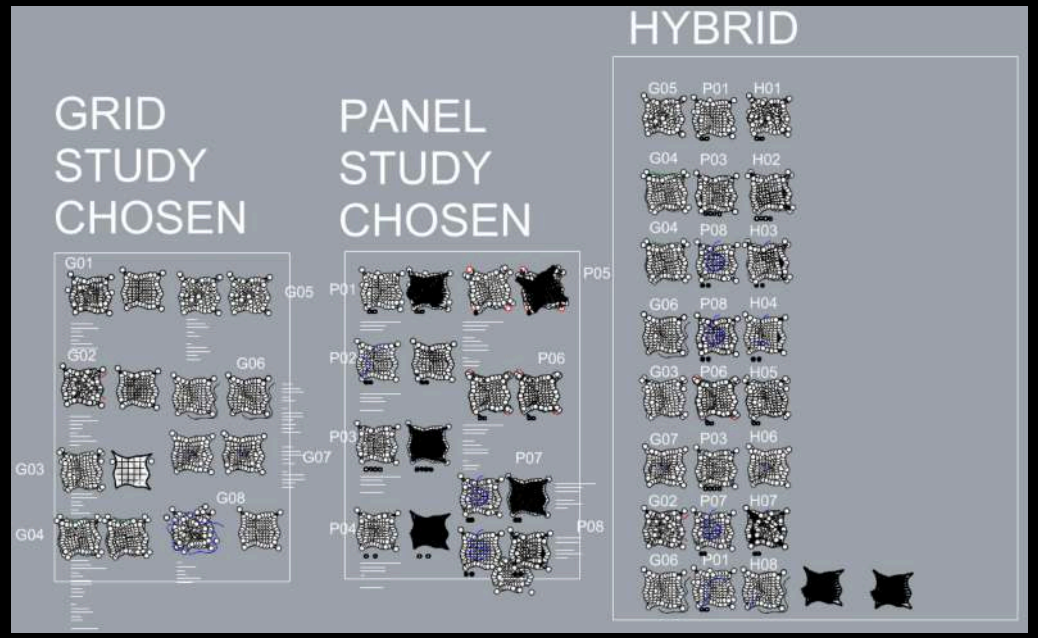
LINKS TO FUTURE PRACTISE

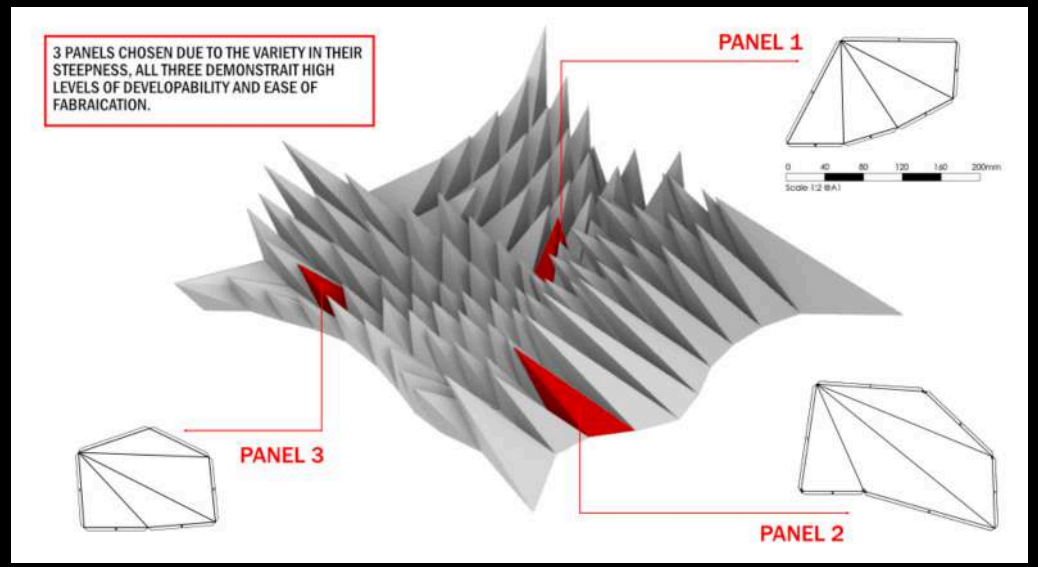
CONVENTION AND ORGANISATION

WITHIN MY DISCIPLINE OF ENGINEERING A CRUCIAL PRACTISE IS FORMULATED AROUND CONVENTIONS AND ORGANISATION. THIS AIMS TO ACHIEVE OPTIMAL COMMUNICATION WITHIN A TEAM, HAVE A CLEAR PATH OF ITERATION AND RETAINS THE ABILITY TO COMMUNICATE TO STAKEHOLDERS. THE USE OF RHINO FOR LABELLING AND RECORDING THE PARAMETRIC DATA IN THIS ASSIGNMENT HAS EDUCATED ME ON THE VITAL ROLL OF CONVENTIONS AND DIGITAL ORGANISATION ON A SMALL SCALE. BY LABELLING THE PARAMETRIC DATA I WAS ABLE TO EASILY ALTER SMALL ASPECTS AND REMOVE THE 'GUESSING GAME' FOR THE BEHAVIOUR OF THE MODELS. I WILL TAKE THIS INTO MY FUTURE PRACTISE AS A VALUABLE LESSON ON STICKING TO A CONVENTION IN ORDER TO ORGANISE AND COMMUNICATE DATA.

FABRICATION AND DEVELOPABILITY

WITH THE RESTRICTION BEING PLACED ON THE MODELS FOR THIS ASSIGNMENT IN TERMS OF DEVELOPABILITY I GATHERED A MORE ACUTE UNDERSTANDING OF DESIGNING AND COMPROMISING FOR FABRICATION AND DEVELOPABILITY. SO MUCH OF THE ENGINEERING SIDE OF DESIGN IS CENTERED AROUND DEVELOPING, MACHINING AND COMMUNICATION WITH FABRICATORS AND IN ORGER TO ACHIEVE THIS THE DESIGNS NEED TO BE DEVELOPABLE. WHILE WORKING IN RHINO ON MY MODEL I UNDERSTOOD THAT LESS UNIQUE AND EXTREME DESIGNS NEEDED TO BE CHOSEN TO EVENTUATE MY HYBRID DUE TO DEVELOPABILITY BEING A CONSTRAINT. I WILL CARRY THIS DESIGN PHILOSOPHY WITH ME INTO MY FUTURE IN ENGINEERING. I WILL ALWAYS STRIVE TO PUSH THE BOUNDS OF DESIGN HOWEVER, THE SKILL TO DESIGN FOR FABRICATION IS ONE I'LL RETURN TO OFTEN.





5 YEAR PLAN TO MASTER COMPUTATIONAL TOOLS

Bachelor of Design

Masters in Engineering

2024

2025

2026

2027

2028 -

INTRODUCTION TO COMPUTATIONAL TOOLS THROUGH THE RESEARCH AND EXPLORATION IN RHINO. GAINING A BASE UNDERSTANDING OF THE MATHEMATICS OF COMPUTATION WITHIN PARAMETRICS THROUGH FIRST YEAR MATHS SUBJECTS.

REFINE AND DEVELOP UNDERSTANDING OF THE PRACTICAL USES OF COMPUTATIONAL DESIGN IN CONJUNCTION WITH FURTHERING KNOWLEDGE OF THE MATHEMATICS BEHIND THE COMPUTATION. THIS WILL BE ACHEIVED THROUGH SUBJECTS SUCH AS ENGINEERING MATHEMATICS AND PROGRESSING THE USE OF SUCH TOOLS WITHIN DESIGN PROCESSES AND TECHNIQUES. ALONG WITH MY OWN REASEARCH AND TIME SPENT UTILISING COMPUTATIONAL TOOLS SUCH AS THOSE IN THIS ASSIGNMENT TO DESIGN AND REALISE CONCEPTS.

ENGINEERING MASTERS CAPSTONE PROJECTS HOLD THE POTENTIAL TO EITHER DEVELOP OR UTILISE COMPUTATIONAL TOOLS. THIS WILL BE A POINT TO REFLECT ON THE ROLE COMPUTATIONAL TOOLS PLAY ON A LARGE SCALE LEADING INTO A LOOK INTO THE FUTURE OF WHERE I WILL USE COMPUTATIONAL DESIGN IN THE WORLD POST UNVERSITY. WHILE STILL AIMING TO MASTER COMPUTIONAL TOOLS.

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Slide 14:

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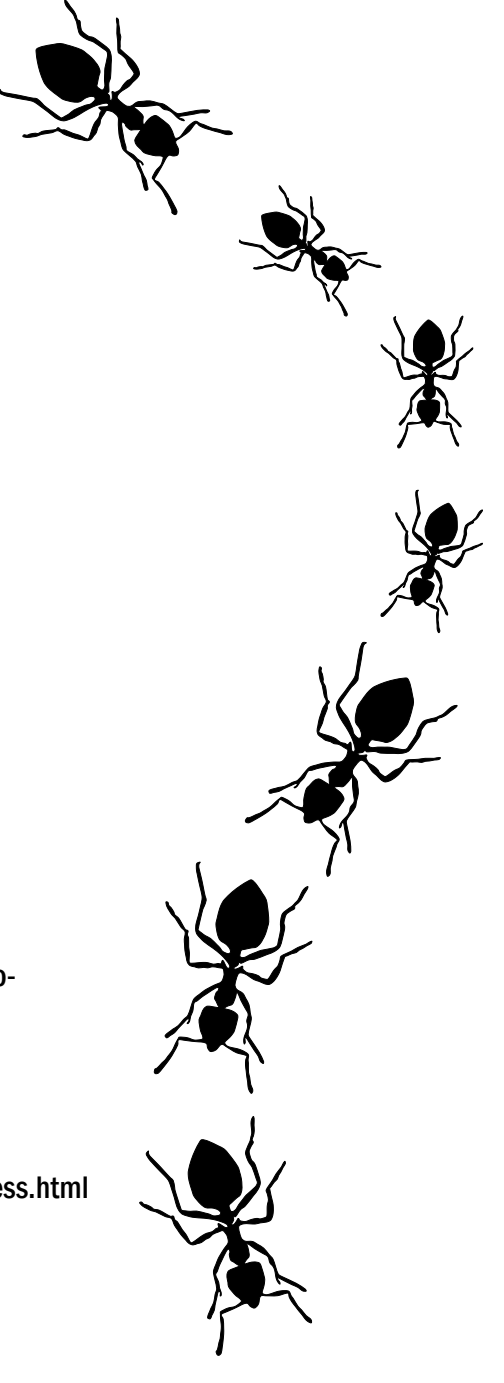
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